

ISLAMIC UNIVERSITY OF TECHNOLOGY

PROJECT REPORT: PHASE A & B (SW)

Team Name: ReVOLT

Team Members:

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Supervisor: Mirza Fuad Adnan

Phase A:

1. Clock Pulse Generator
2. Program Counter
3. Input Unit and MAR

Phase B: Memory Units

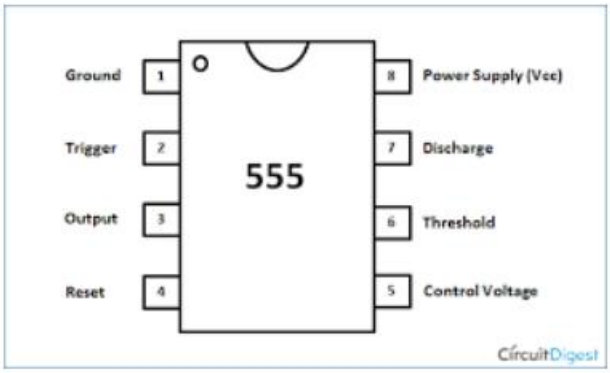
1. Registers
    - Accumulator
    - B Register
    - Output Register
2. Random Access Memory (RAM)

Phase A

1. Clock Pulse Generator:

This provides the required clock pulse signal to all other sub modules to ensure the full functionality of the overall system.

Clock pulse would drive all the sequential circuits that we used to implement the computer. For generating it we need to use 555 timer ic.



This is the pin diagram of the 555 Timer IC. The IC should have a power supply at pin-8 to provide Vcc and pin-1 should be grounded.

Clock pulse generator have two modes of operation.

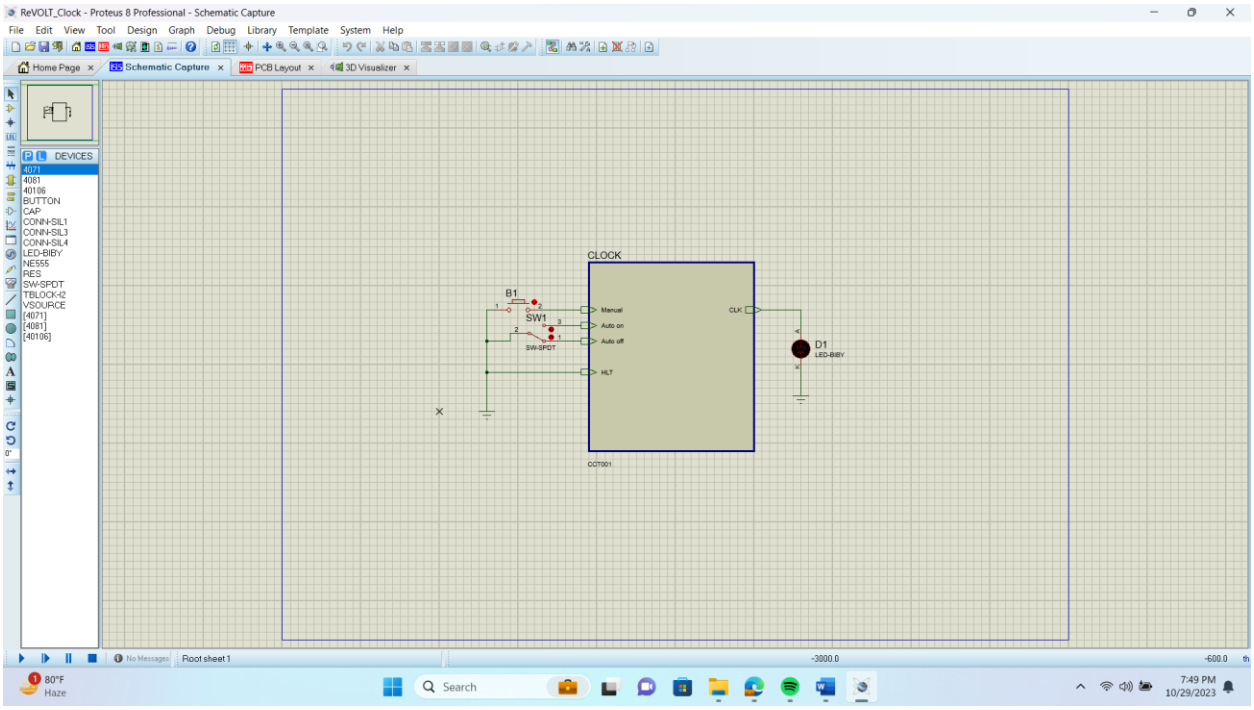
a) Automatic Mode: In this case the generated signal would be continuous identical to the output of an astable multivibrator.

b) Manual Mode: In this case the clock pulse will have a low output unless the operator manually instructs the clock generator to generate a single pulse, a mode of operation that is similar to a monostable multivibrator.

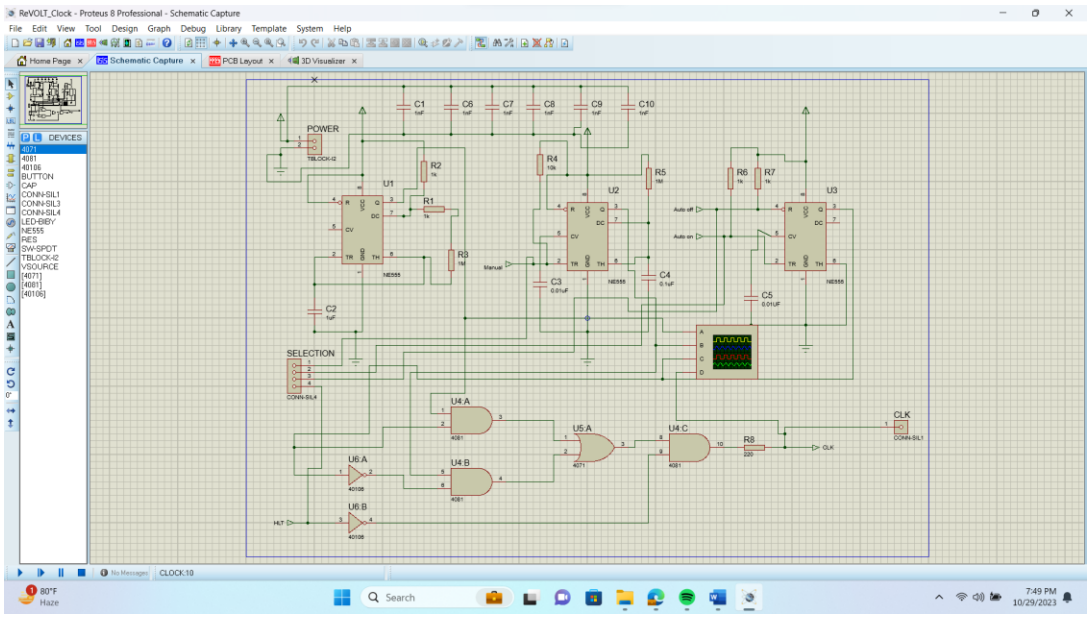
Software Implementation:

In the parent circuit the clock generator has select pin which is used to choose between automatic mode or manual mode of the clock pulse.

When the switch is up it will be used as manual mode. When the push button is pressed a single clock pulse is generated. When the switch is down it will be used as automatic mode.

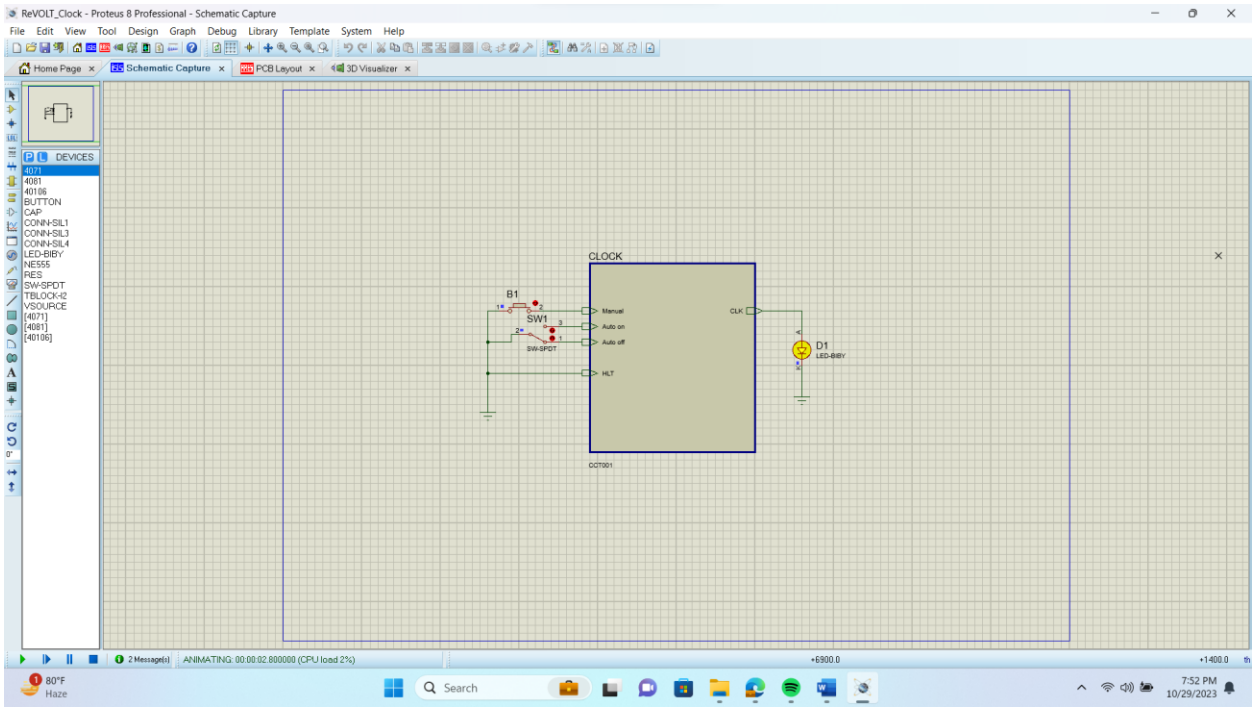


In the subcircuit, there are three 555 timer ICs. The first one is used as an astable multivibrator, the second one is used as a monostable multivibrator and the third one is used as a bistable multivibrator. 555 timer ICs pin 8 is connected with source and pin 1 is connected with ground. After that all the 555 timer ICs connected with logic gates. We used 74HC04, 74HC08 and 74HC32 in this regard. Pin 7 of the logic gates ICs is connected with ground and pin 14 I connected with voltage source. We need to ground the HLT input so that it will be 1 after passing the NOT gate.

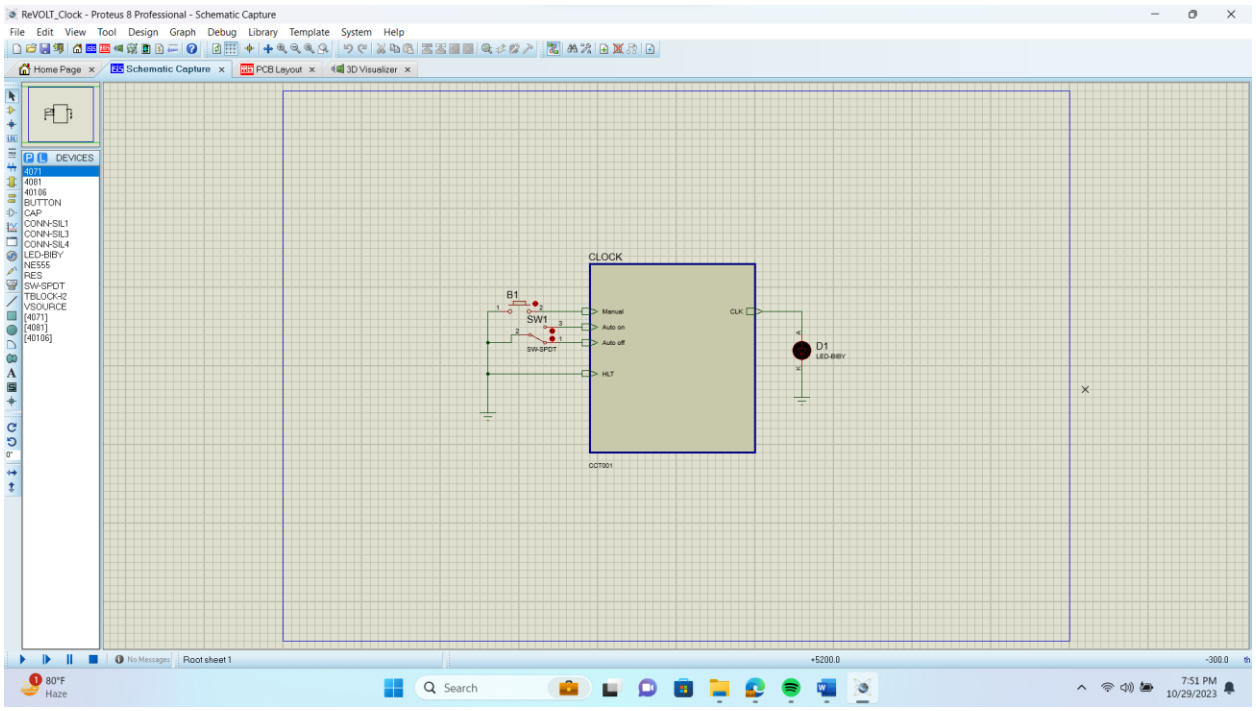


output

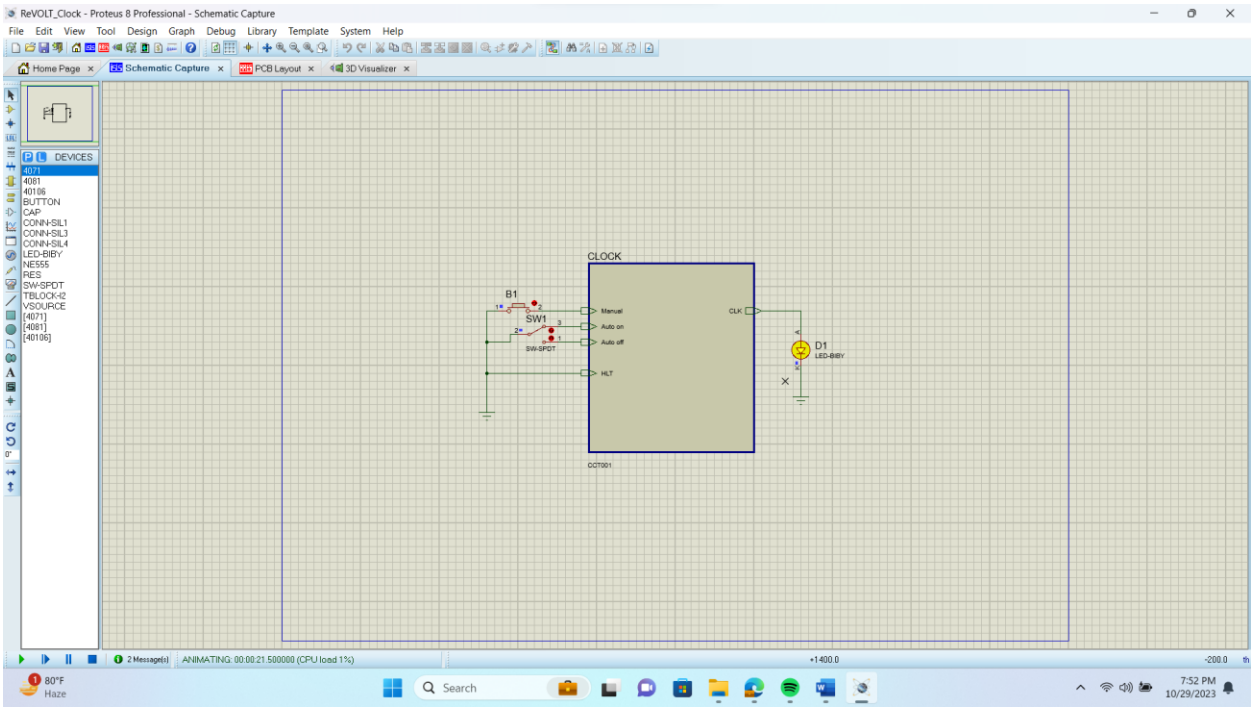
When the switch is down(manual mode on) & push button is pressed:

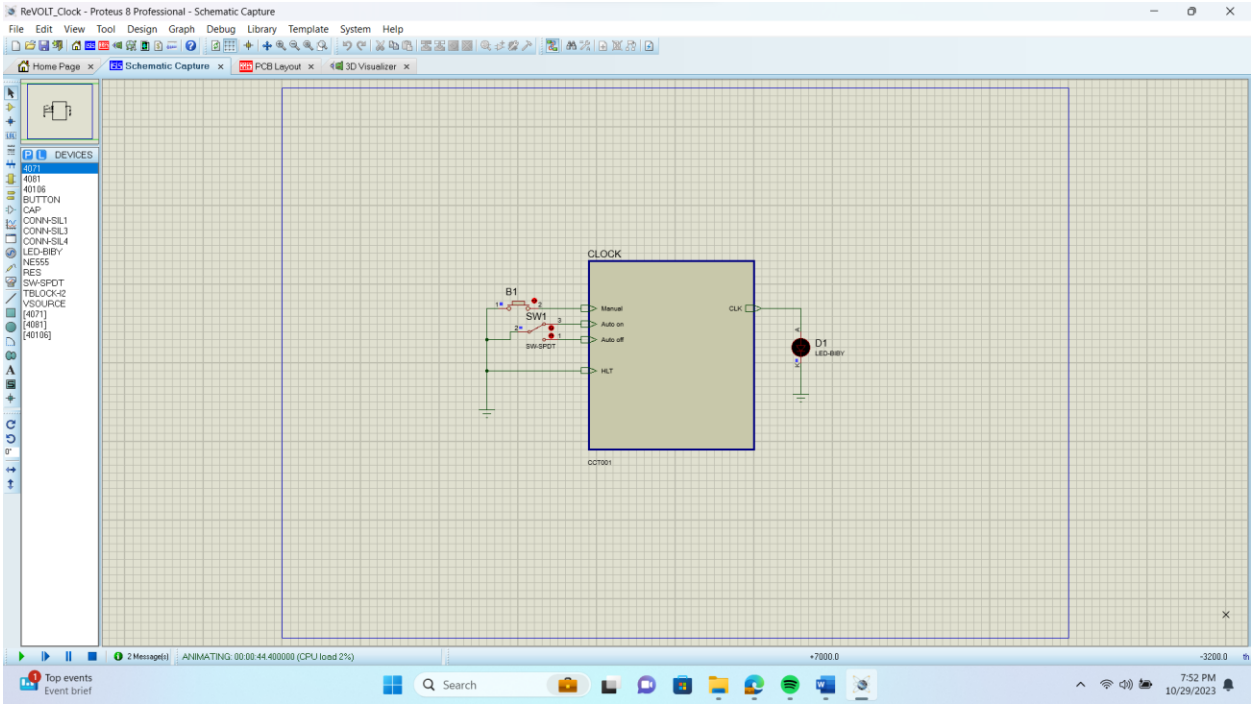


When the switch is down(manual mode on) & push button is open:

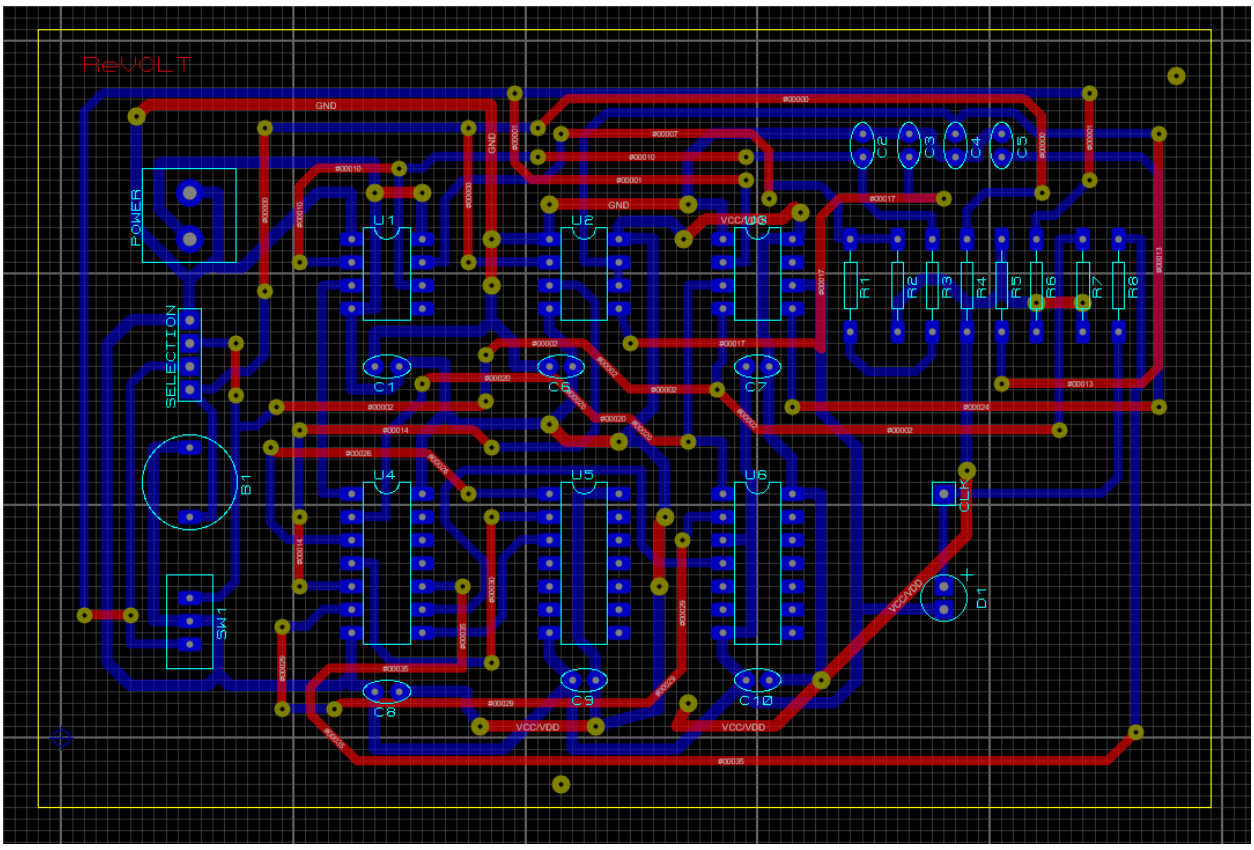


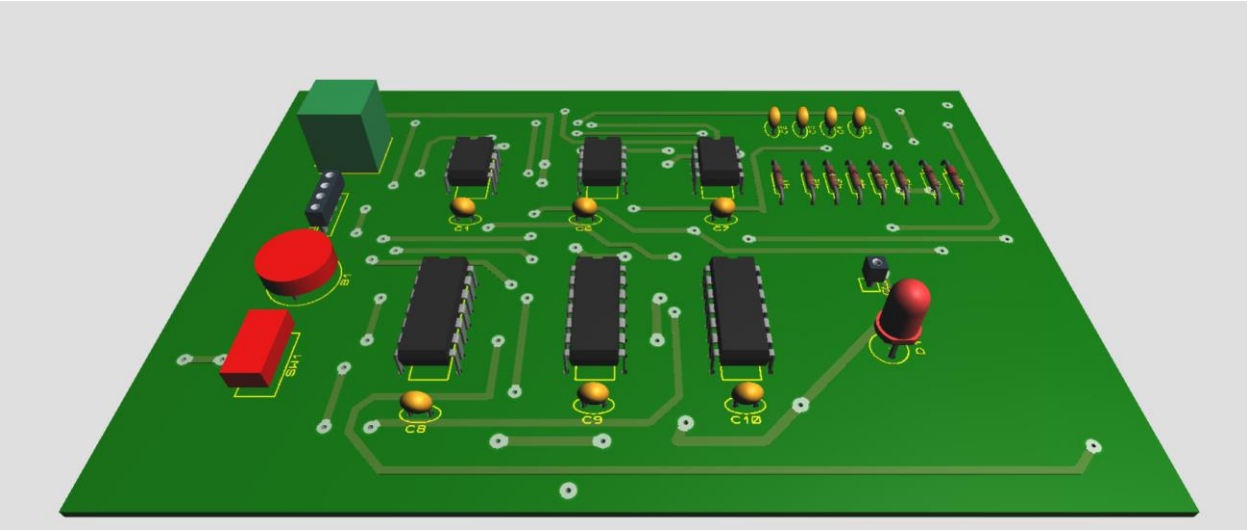
When the switch is down i.e, auto is on(continously led blinking):




ReVOLT_Clock - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Home Page
Schematic Capture
PCB Layout
3D Visualizer
DEVICES
4071
4081
40106
BUTTON
CAP
CONN-SIL1
CONN-SIL3
CONN-SIL4
LED-BIBY
NE555
RES
SW-SPDT
TBLOCK-I2
VSOURCE
[4071]
[4081]
[40106]
CLOCK
B1
SW1
SW-SPDT
Manual
Auto on
Auto off
HLT
CLK
D1
LED-BIBY
CCT001
2 Message(s)
ANIMATING: 00:00:44.400000 (CPU load 2%)
+7000.0
-3200.0
Top events
Event brief
Search
7:52 PM
10/29/2023

PCB Layout:


ReVOLT
GND
POWER
SELECTION
B1
SW1
U1
U2
U4
U5
U6
C1
C2
C3
C4
C5
C7
C8
C9
C10
R1
R2
R3
R4
R5
R6
R7
R8
D1
CLK
VCC/VDD


C7
C8
C9
C10
SW1

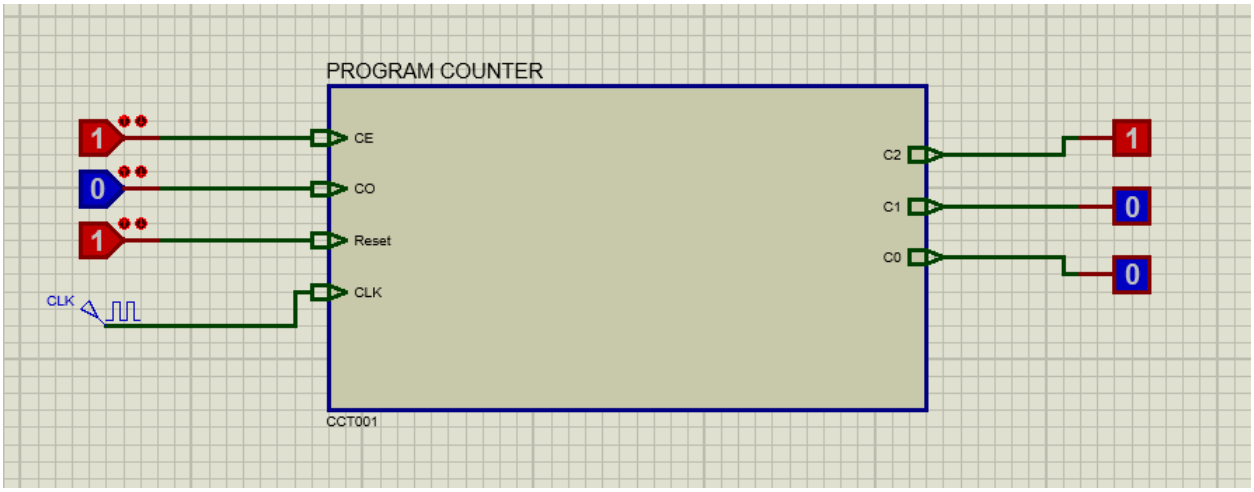
2. Program Counter:

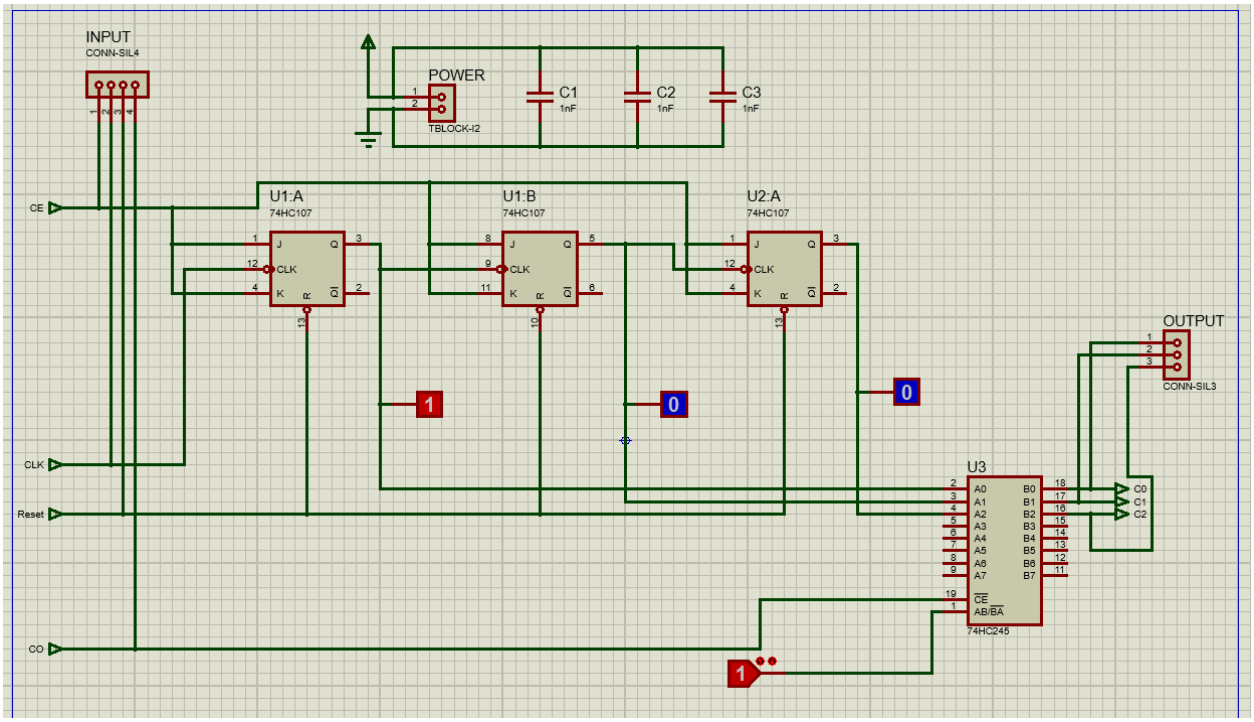
Program counter (PC) is a 3-bit counter which can count from 000 up to 111. The job of a program counter is to keep and send memory addresses for the next instruction which is to be fetched and executed. After each cycle of instruction has been completed, the value will be incremented.

Function of the I/O pins –

Pin	Type	Function
CE	IN	Program Counter Enable: Increments the program counter on the next clock-cycle.
CO	IN	Program Counter Out: Puts the current program counter value on the bus. Output is disconnected when CO is low.
RESET	IN	Resets the program counter to 000.
CLK	IN	Clock pulse input.
OUT (C0, C1, C2)	OUT	3-bit counter output.

Schematic Layout:





Design:

A 3-bit asynchronous counter was designed in order to implement program counter. Here JK flip flop ICs and octal 3−state noninverting bus transceiver IC were used. The first flip flop is clocked by external pulse and then each successive flip flop is clocked by Q output of previous stage. As soon the clock of second stage is triggered by output of first stage, the second stage gives output. Because of inherent propagation delay time all flip flops are not activated at same time which results in asynchronous operation.

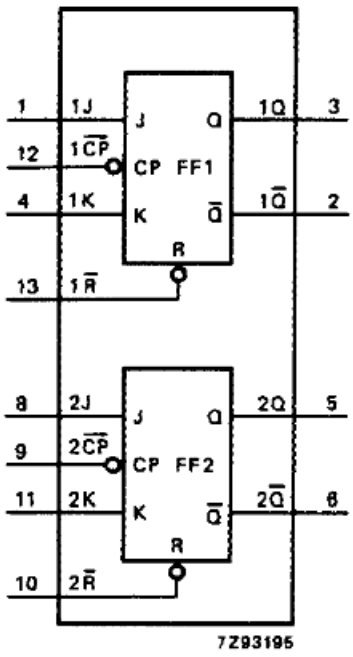
Description of used ICs:

a) IC 74HC107 – Dual JK Flip-Flop IC with Reset:

The 74HC107 is a dual negative edge triggered JK flip-flop featuring individual J and K inputs, clock and reset inputs and complementary Q

and Q outputs. The reset is an asynchronous active LOW input and operates independently of the clock input. The J and K inputs control the state changes of the flip-flops as described in the mode select function table. The J and K inputs must be stable one set-up time prior to the HIGH-to-LOW clock transition for predictable operation.

The input CE is connected to J & K and Reset and Clock are connected to the pins according to the diagram –



Functional diagram.

b) IC 74HC245 – Octal 3−state noninverting bus transceiver:

It is a 20-pin IC, the detail of all pins is mentioned below.

Pin#01 DIRECTION CONTROL PIN: The direction control pin (DIR) is used to decide the direction of communication whether data will from transport A to B or from B to A.

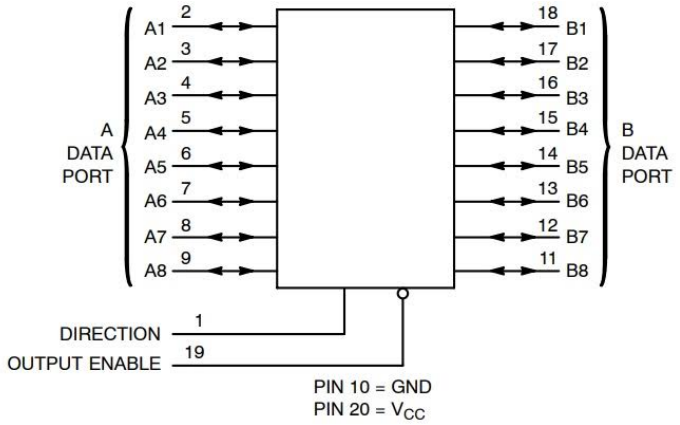
Pin#02 to 09 A0-A7: These are input pins (as used here) that also behaves as output depending on the direction control input.

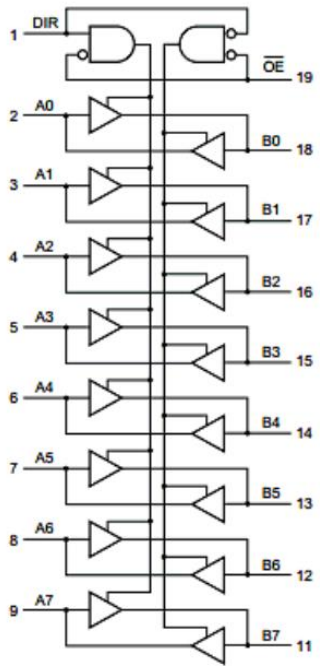
Pin#10 Ground

Pin#11 to 18 B0-B7: These are output pins (as used here) that also behaves as input depending on the direction control input.

Pin#19: It is the active-low output enable pin. It controls the output by enabling or disabling the inputs. This pin is used for cascading.

In the pin 1, always 1 is given to denote that data is going from A data port to B data port. The CO and other inputs and outputs are connected to the pins according to the diagrams –





INPUTS:

CO is active low. When CE=1 the flip flops are in toggling condition and when CE=0 the flip flops are in memory condition.

CE is active high. As the CO pin is in active low the outputs will be activated if the CO pin is logic condition 0. If the CO pin is in logic condition 1, the outputs from the bidirectional buffer will not be activated.

Reset is active low. So if reset=0, the counter resets to initial value 000.

Clock pulse is used for CLK input.

OUTPUTS:

3 outputs C0, C1, C2 where C2 is the MSB and C0 is the LSB.

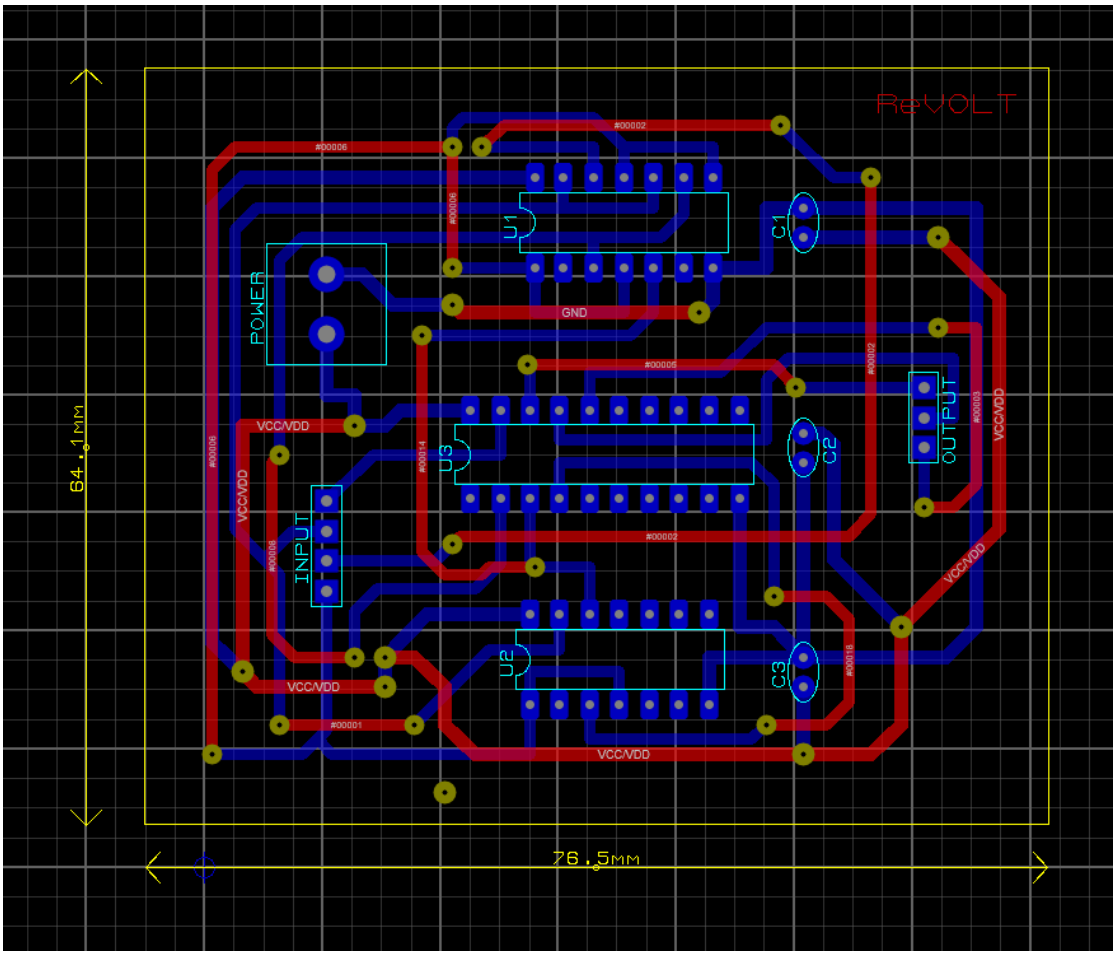
Operation:

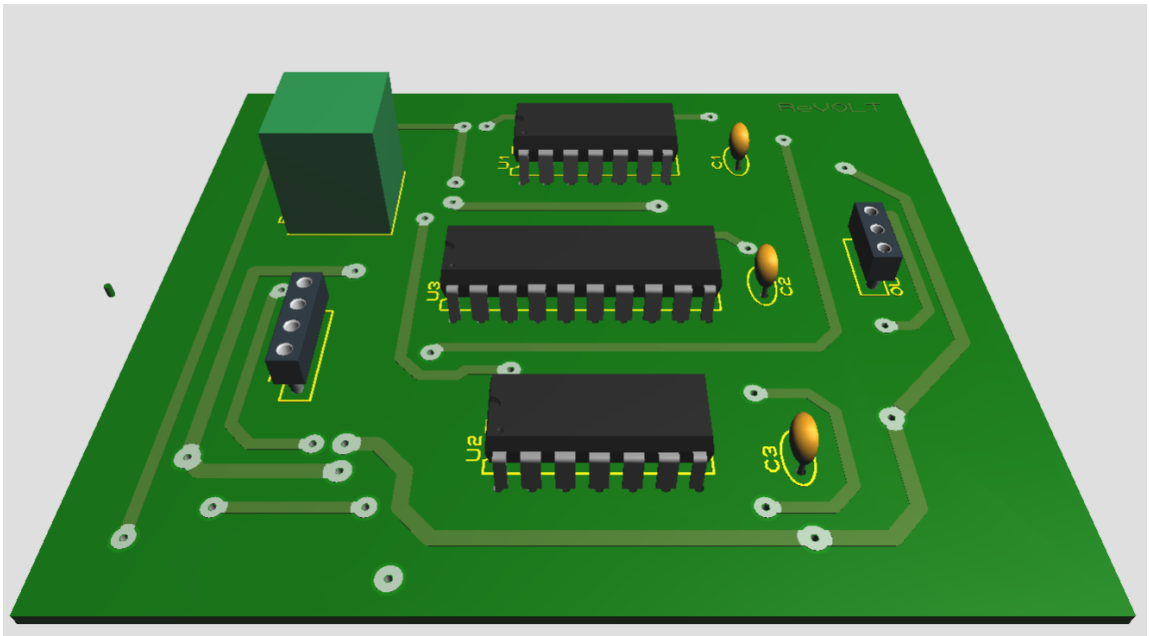
Before simulating, the counter is reset to 000 using the reset pin. When the clock provides a negative edge the instruction cycle begins. Given that, CE=1, the flip flops are set to toggle condition and it toggles the LSB from 0 to 1. As the flip flops are clocked from preceding flip flop. The other flip flops are not activated. During the next cycle, the LSB toggles from 1 to 0 and provides negative edge to the next flip flop. Thus the process goes on and the counter counts from 000 to 111.

Truth table:

Counter State	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

PCB Layout:

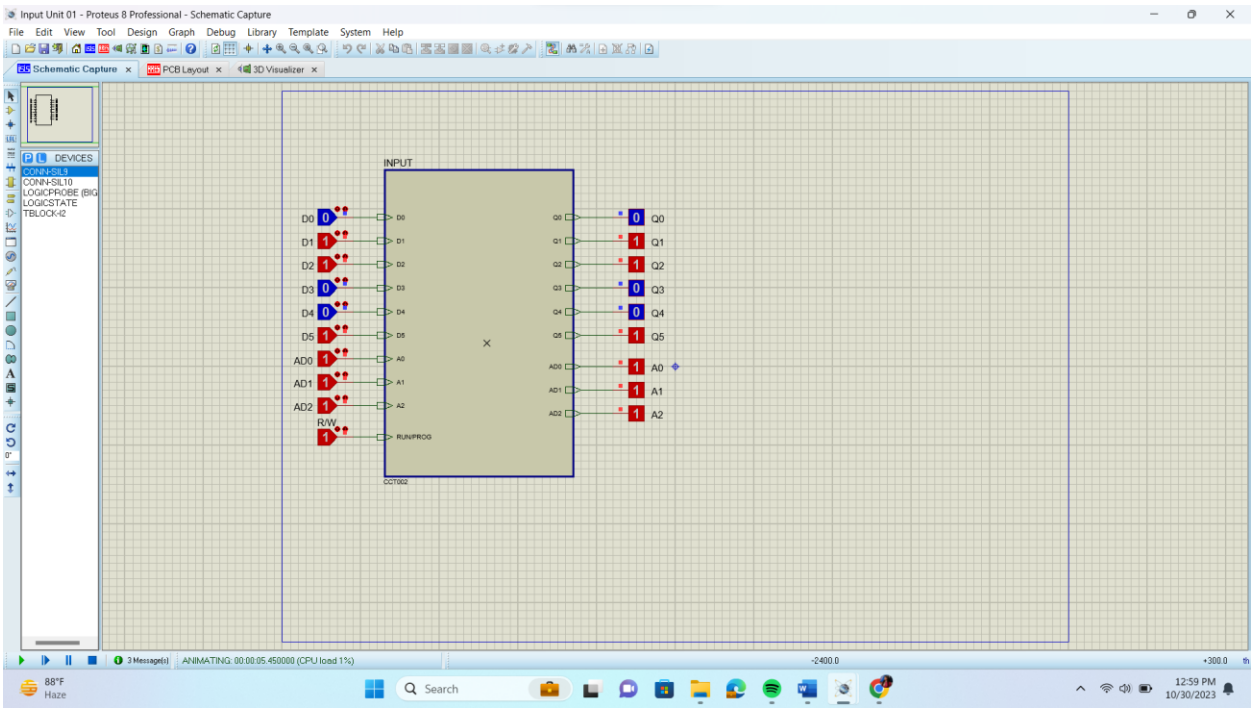

ReVOLT
POWER
INPUT
OUTPUT
GND
VCC/VDD
U1
U2
U3
C1
C2
C3
64.1mm
76.5mm

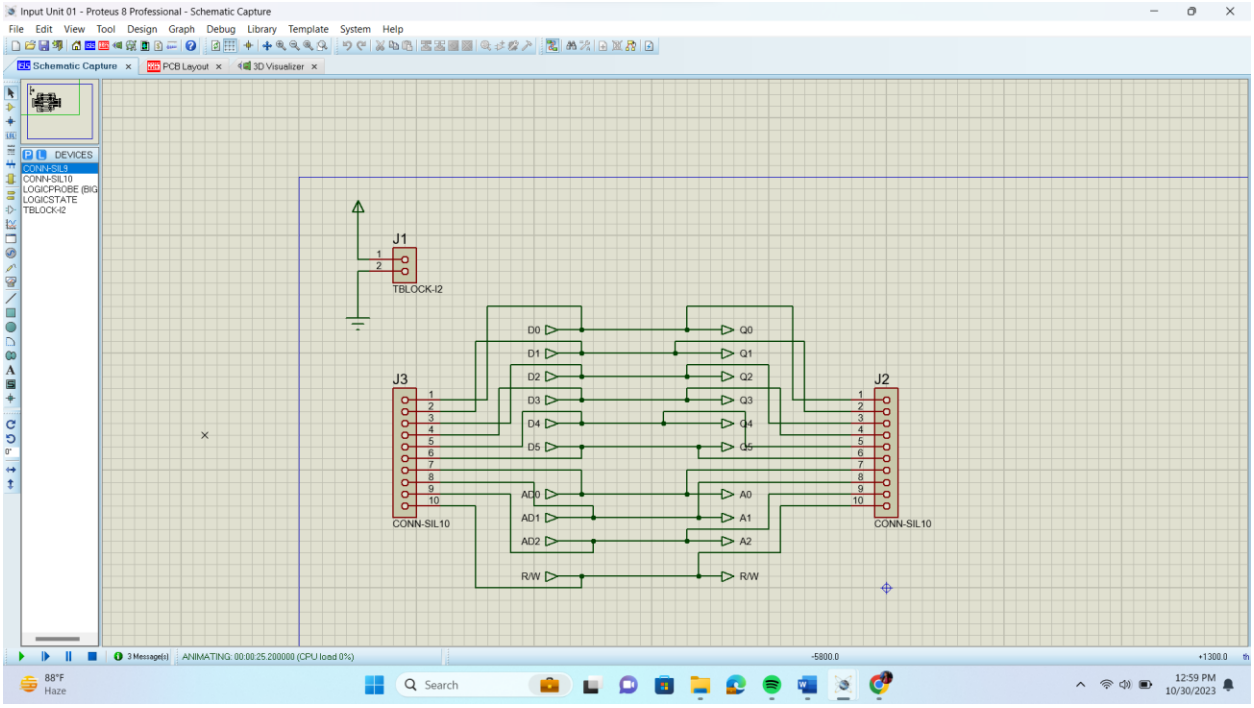

ReVOLT
U1
U2
U3
C1
C2
C3

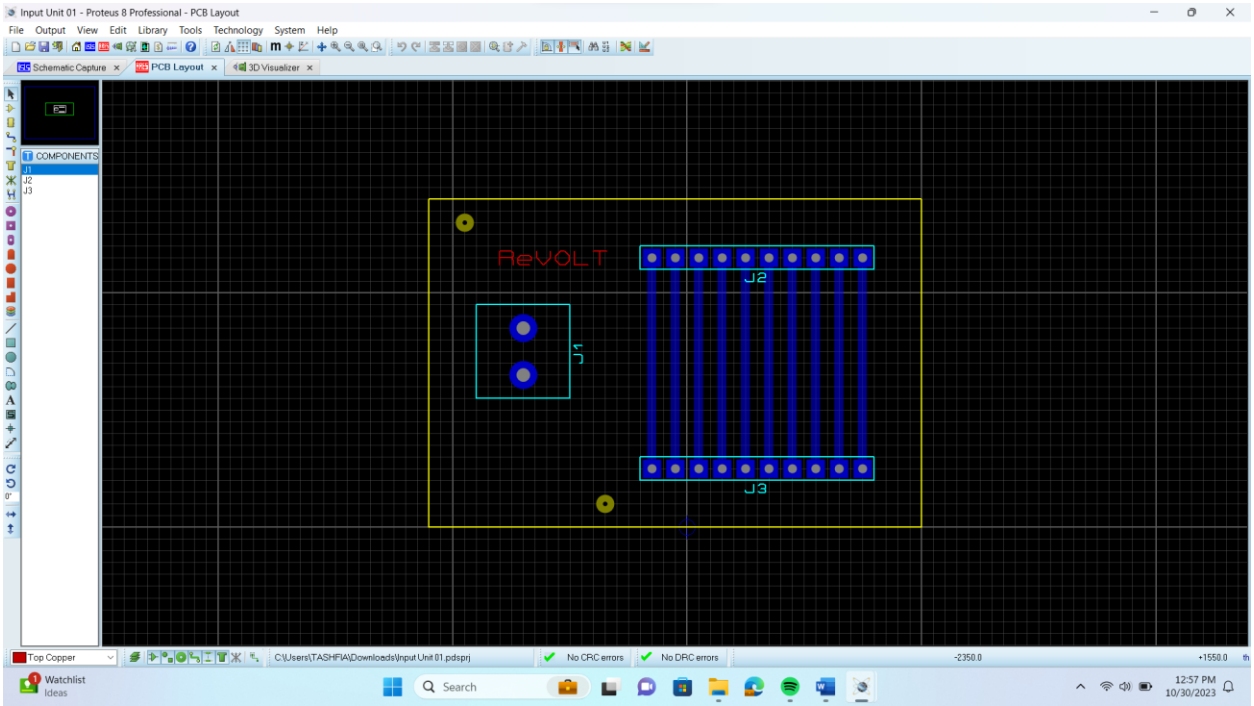
3. Input Unit and MAR:

Input unit

The input unit is a module connected to the RAM and address selctors. When the system is in PROG state (selected in the address selector), input 6 bits are provided to the RAM from input unit. At the same time, Address bits send the information of the address where the input data needs to be stored. The R/W switch of the input unit decides whether data is to be read by the RAM or written into it. The switches in the Inout unit are respectively: 1. 6 switches that regulate the input data 2. 3 switches that regulate address bit 3. 1 R/W switch to regulate Read or Write mode in RAM.




Input Unit 01 - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
PCB Layout
3D Visualizer
DEVICES
CONN-SIL9
CONN-SIL10
LOGICPROBE (BIG
LOGICSTATE
TBLOCK-I2
J1
TBLOCK-I2
J3
CONN-SIL10
J2
CONN-SIL10
D0
D1
D2
D3
D4
D5
Q0
Q1
Q2
Q3
Q4
Q5
AD0
AD1
AD2
A0
A1
A2
R/W
R/W
3 Message(s)
ANIMATING: 00:00:25.200000 (CPU load 0%)
-5800.0
+1300.0
88°F
Haze
Search
12:59 PM
10/30/2023

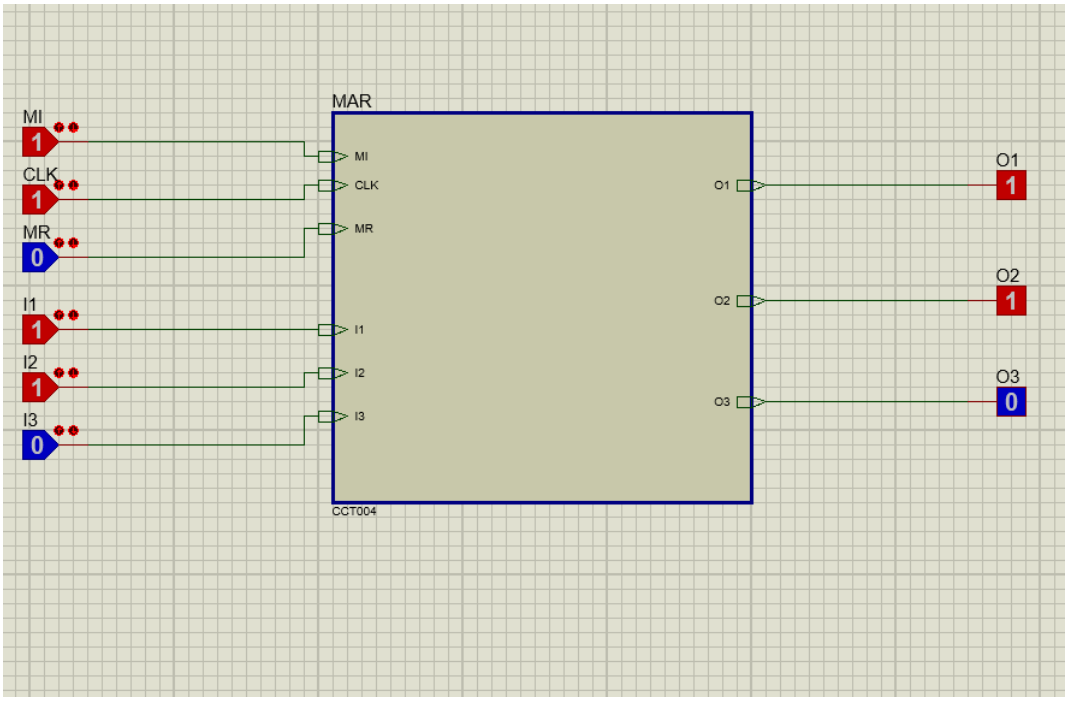

Input Unit 01 - Proteus 8 Professional - PCB Layout
File Output View Edit Library Tools Technology System Help
Schematic Capture
PCB Layout
3D Visualizer
COMPONENTS
J1
J2
J3
ReVOLT
J1
J2
J3
Top Copper
C:\Users\TASHFIA\Downloads\Input Unit 01.pdsprj
No CRC errors
No DRC errors
-2350.0
+1550.0
Watchlist
Ideas
Search
12:57 PM
10/30/2023

Memory Address Register (MAR)

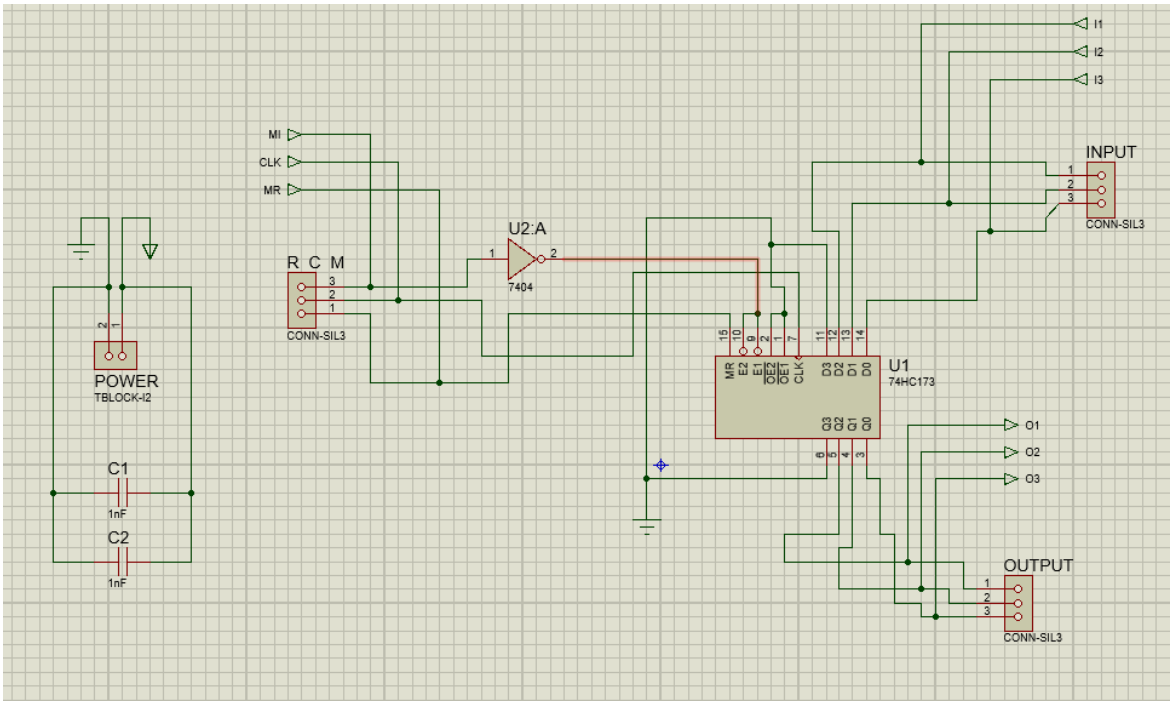
MAR or memory address register is a register which is responsible for storing memory address of the data . The MAR holds the location in memory where data or instructions are read .When the computer is in RUN state , the address is obtained from the counter through the bus and then stored.

There we may find some pins which do the storing process . Like MI , CLK , RESET , INPUT and OUTPUT .

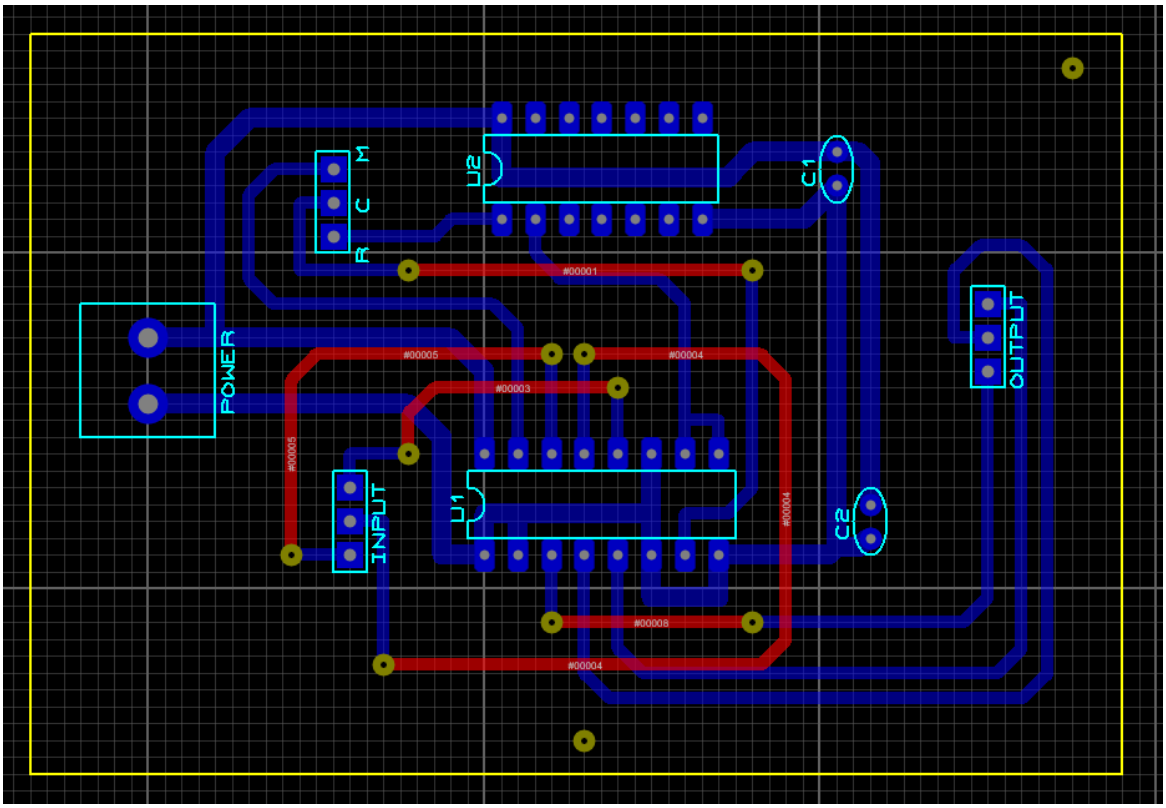
CIRCUIT DIAGRAM:



SUB CIRCUIT OF MAR



CHILD SHEET OF MAR



PCB OF MAR

Here IC- 74HC173 is used for the MAR

PIN DESCRIPTION :

MI: It is a pin which works as an enable input of the MAR . We use 9th and 10th pin of the IC to make the MI input connection line . This 9th and 10th pin is named as E1 and E2 . When MI is on then MAR starts to take address . Otherwise it will show previous value.

CLK: It is a clock pulse generator which provides pulse input . It is connected to pin no 7.

I**NPUT** : There for a 6 bit SAP 3 input is needed .So MSB is grounded and the remaining inputs are taken . Pin no 12,13 and 14 are connected as input pin.

OUTPUT : Here 3 outputs are taken from the pin 3,4 and 5. The MSB is connected with the ground .

RESET: There is a specific pin for the Reset process . Pin no 15 is used as reset pin .

At last 1,2 ,11 and 6 are grounded .

OPERATION OF MAR

When the MI pin is high, then for CLK high, the inputs given in the MAR will be shown at the output. If the MI pin is then made low, no matter how the inputs are changed, the output will remain the same as the first case. If the RESET pin is made high then the MAR resets totally. If the MI pin is again made high, then change in input will cause a change in the output again.

Phase B

1. Registers

- Accumulator

- B Register

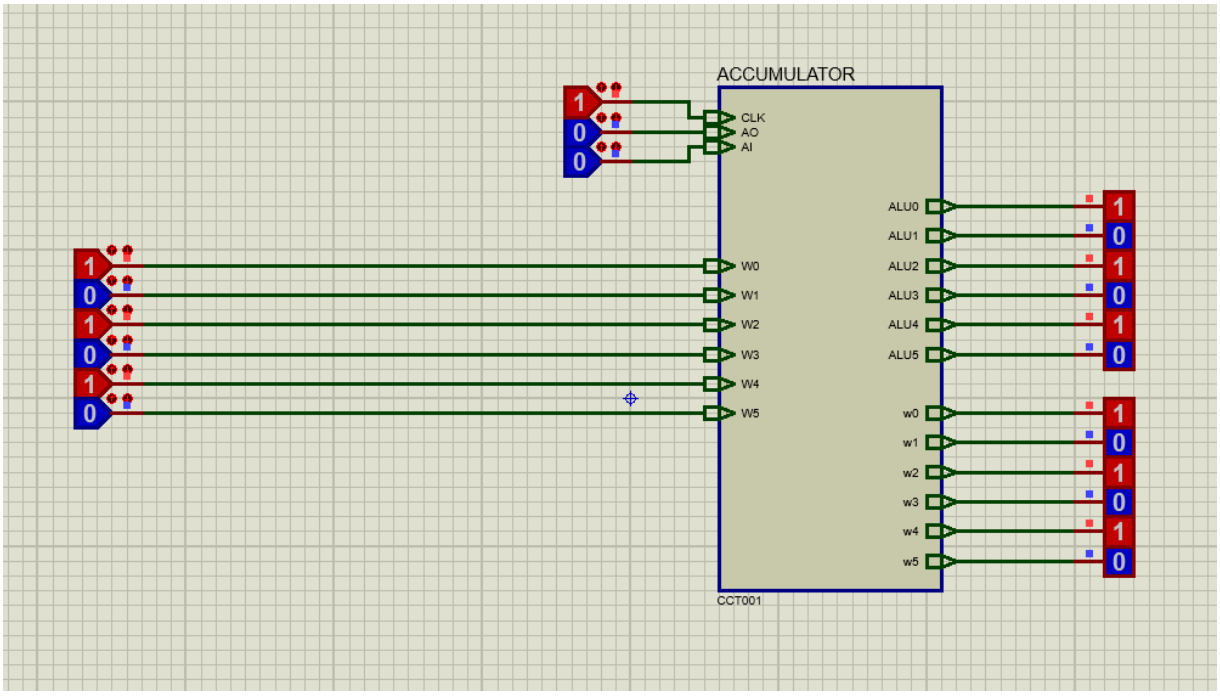
- Output Register

All of the registers are of 6-bit size. Accumulator (also called as A register) should have the ability to output intermediate results after each ALU operation. Accumulator and B register are used to store 6-bit data as input to ALU input. The purpose of the output register is to store result to display when requested. Controller/sequencer unit sends control signals to these registers for controlling input/output enable. Description of the control bits –

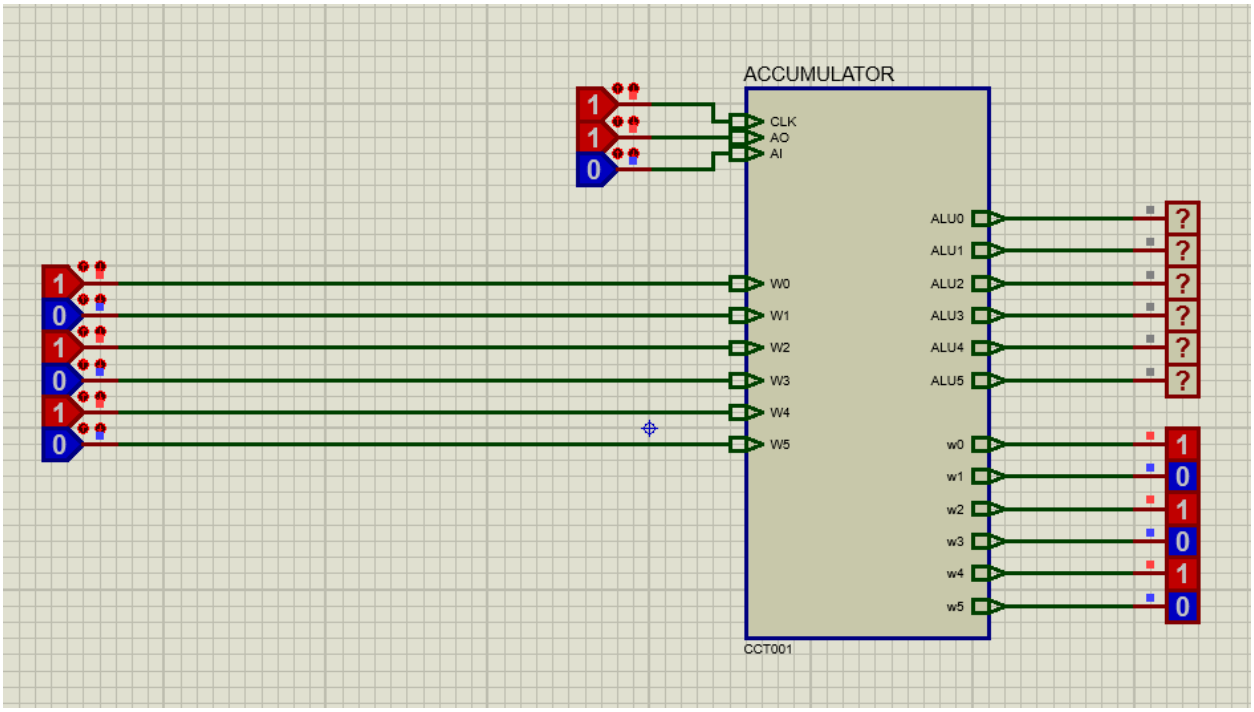
PIN	TYPE	FUNCTION
AI	Accumulator In	Stores the current values of the bus into Accumulator.
AO	Accumulator Out	Accumulator sends its stored content to the bus. Output of accumulator is disconnected when AO is low.
BI	B Register In	Stores the current values of the bus into B Register.
OI	Output Register In	Stores the current values of the bus into Output Register.

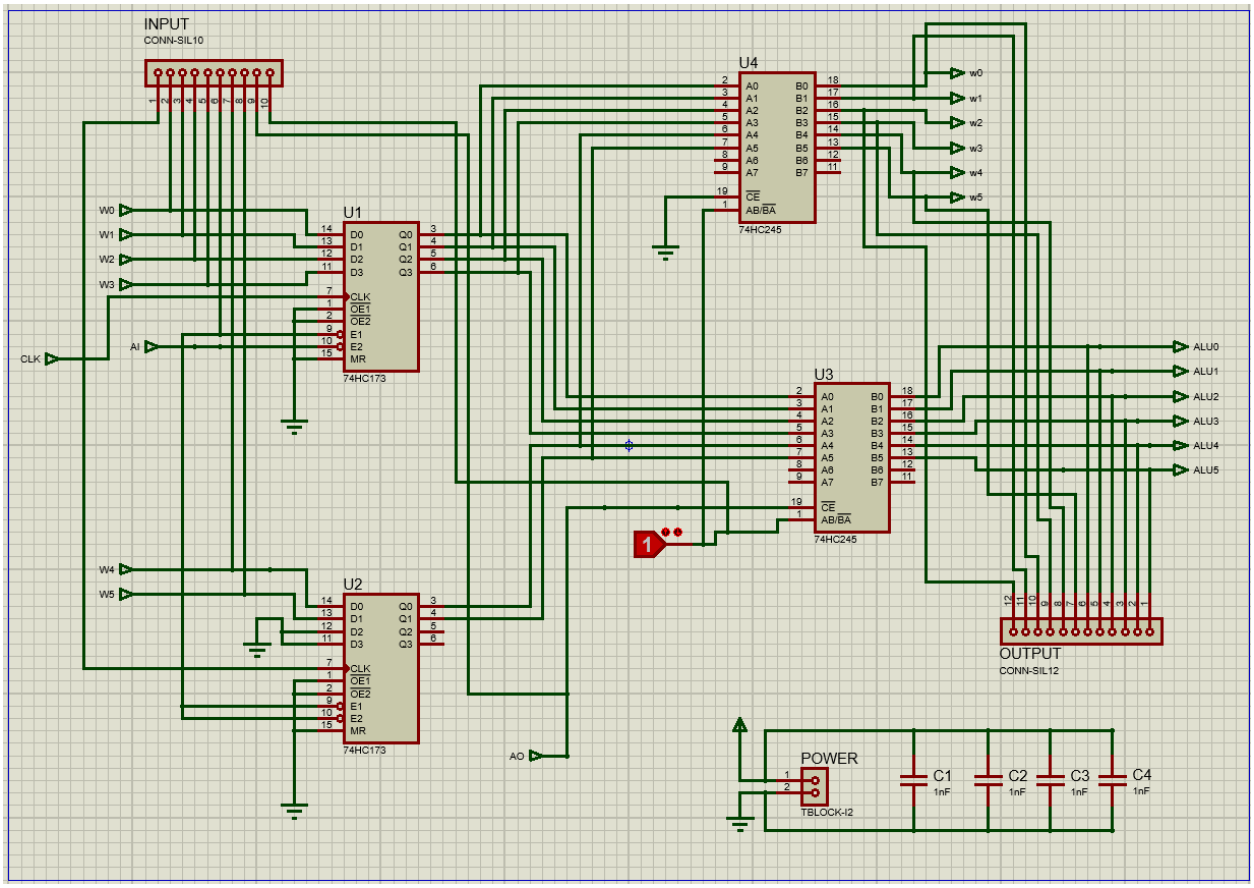
Accumulator:

Schematic Layout:



With AO=1:

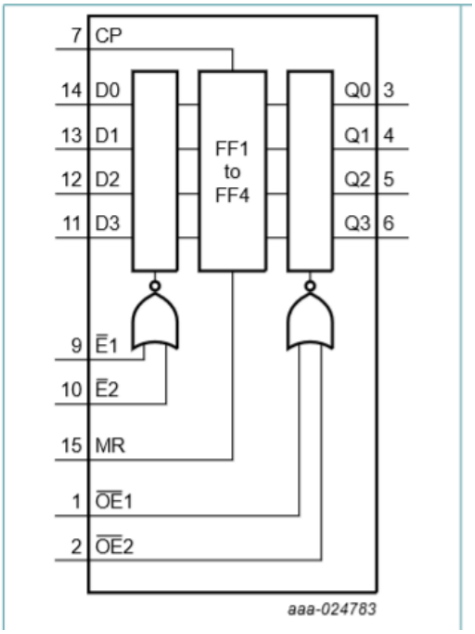




Description of used ICs:

a) IC 74HC173 – Quad D-type flip-flop positive-edge trigger:

The device features clock (CP), master reset (MR), two input enable (E1, E2) and two output enable (OE1', OE2') inputs. When the input enables are LOW, the outputs Qn will assume the state of their corresponding Dn inputs that meet the set-up and hold time requirements on the LOW-to-HIGH clock (CP) transition. A HIGH on either input enable will cause the device to go into a hold mode, outputs halt their previous state independently of clock and data inputs. A HIGH on MR forces the outputs LOW independently of clock and data inputs. Operation of the output enable inputs does not affect the state of the flip-flops.



b) IC 74HC245 – Octal 3−state noninverting bus transceiver:

Described earlier.

Design:

There are two 74HC173 ICs which works as 4-bit D registers. Each contains 4 input data pins as well as 4 output pins. 4 inputs are given and 4 outputs are gotten from the first IC. And 2 inputs are given and 2 outputs are gotten from the second IC. Thus 6 inputs are given and 6 outputs are gotten from the 74HC173 ICs.

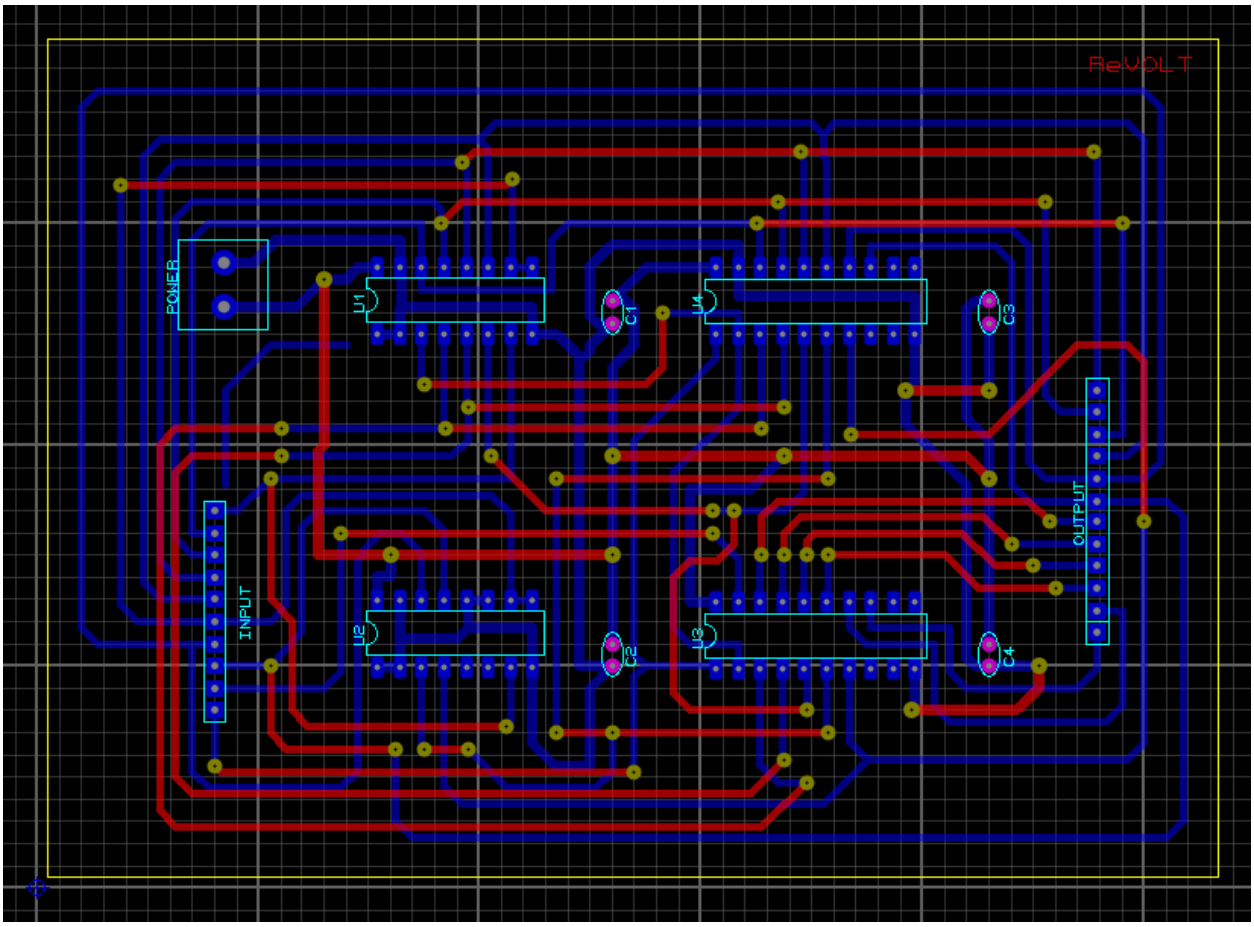
The 6 outputs are passed through 2 74HC245, bus transceiver, making total of 12 outputs. 6 of the outputs goes to the bus which are w0, w1, w2, w3, w4, w5 and 6 of the outputs goes to the ALU which are ALU0, ALU1, ALU2, ALU3, ALU4, ALU5.

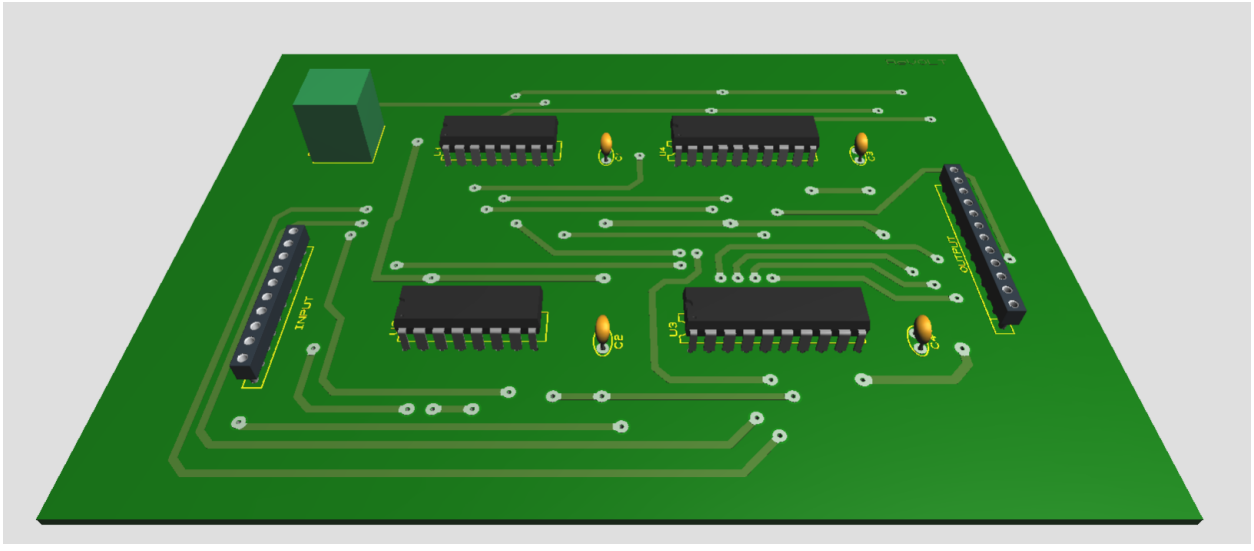
In case of the first 74HC245, which is connected to the outputs which go to the bus, the output enable pin (19) is connected to the ground. As it is an active low pin, this configuration ensures that the input always goes to the output.

In case of the second 74HC245, which is connected to the outputs which go to the ALU, the output enable pin (19) is connected AO control bit. As it is an active low pin, when AO=0, output passes and when AO=1, we don't get any output.

In case of the output that goes to the bus, we will get the output we give in the input. In case of the output that goes to the ALU, the output will depend on AI, AO & CLK. We will get the output according to the input in the positive triggers, which means when CLK goes from 0 to 1. If AI=0, we get the output based on input and if AI=1, we get the output in memory.

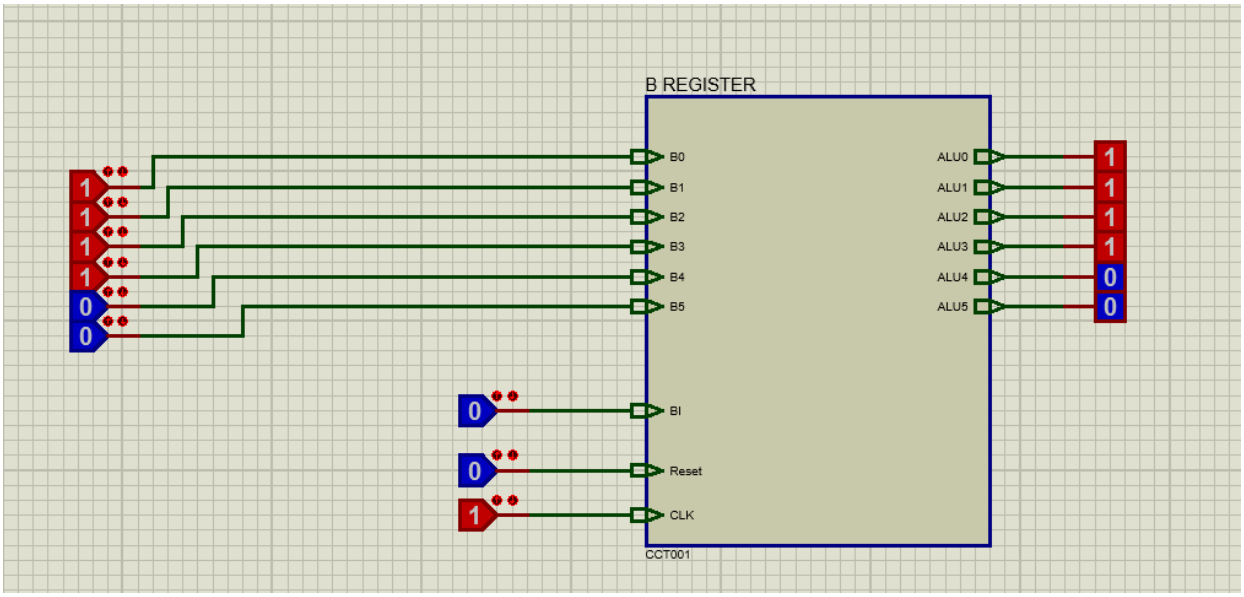
PCB Layout:

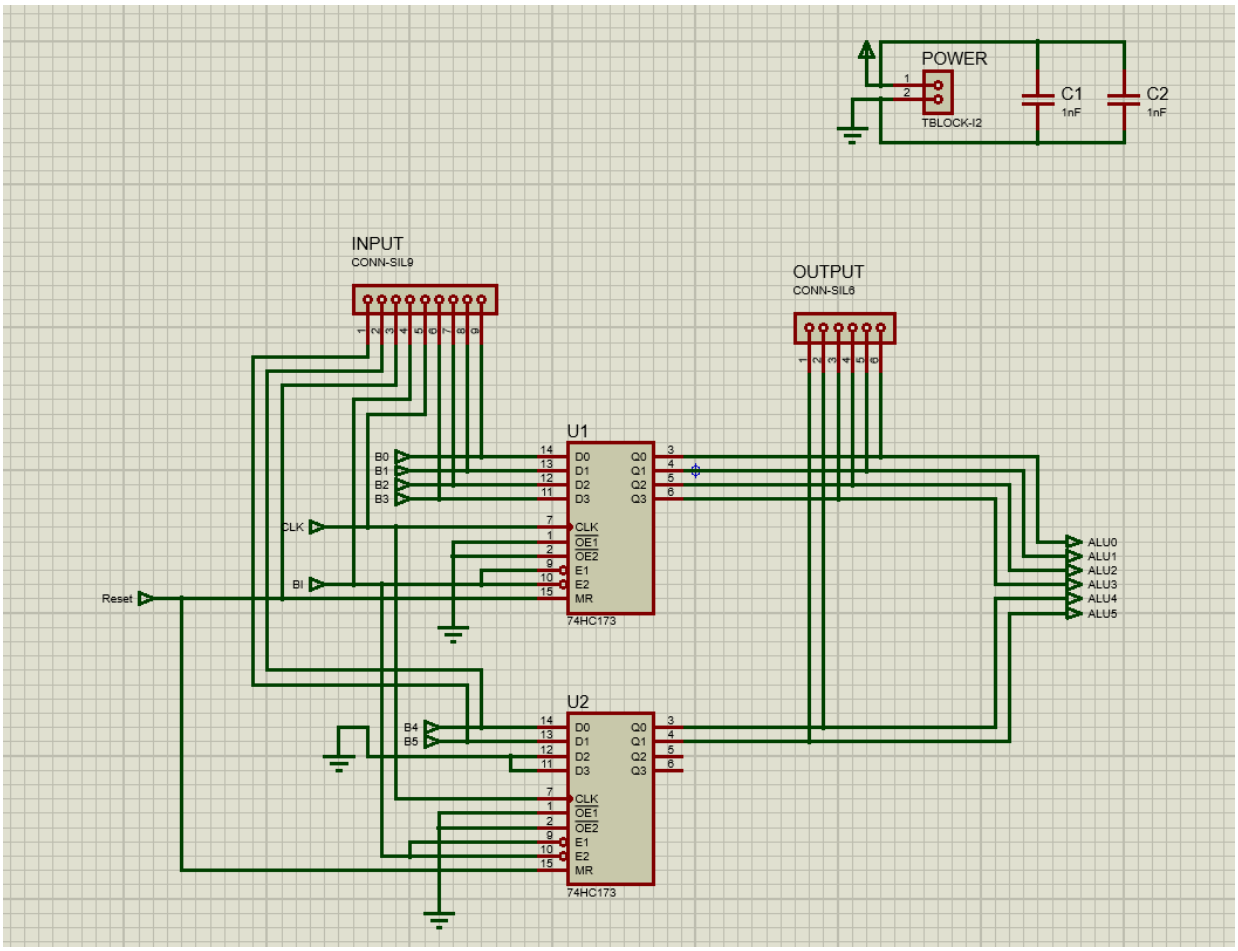

ReVOLT
POWER
U1
C1
U4
C3
INPUT
OUTPUT
U2
C2
U3
C4


INPUT
OUTPUT
C2
U3

B Register:

Schematic Layout:





Design:

There are two 74HC173 ICs which works as 4-bit D registers. Each contains 4 input data pins as well as 4 output pins. 4 inputs are given and 4 outputs are gotten from the first IC. And 2 inputs are given and 2 outputs are gotten from the second IC. Thus 6 inputs are given and 6 outputs are gotten from the ICs.

Input Pins:

BI is active low. When BI=1 the B reg doesn't take input and output shows the memory and when BI=0 the output changes with the clock function.

Reset is active high. So if reset=1, the outputs resets to initial value 0.

When the clock turns from 0 to 1, the output is changed according to the input, because of positive edge triggering.

Whether the data will go to output side from the input side in the IC is dependent on the OE1', OE2' pins. These are active low pins, so the were grounded to always pass the data in the IC.

And 6 inputs B0, B1, B2, B3, B4, B5.

Output Pins:

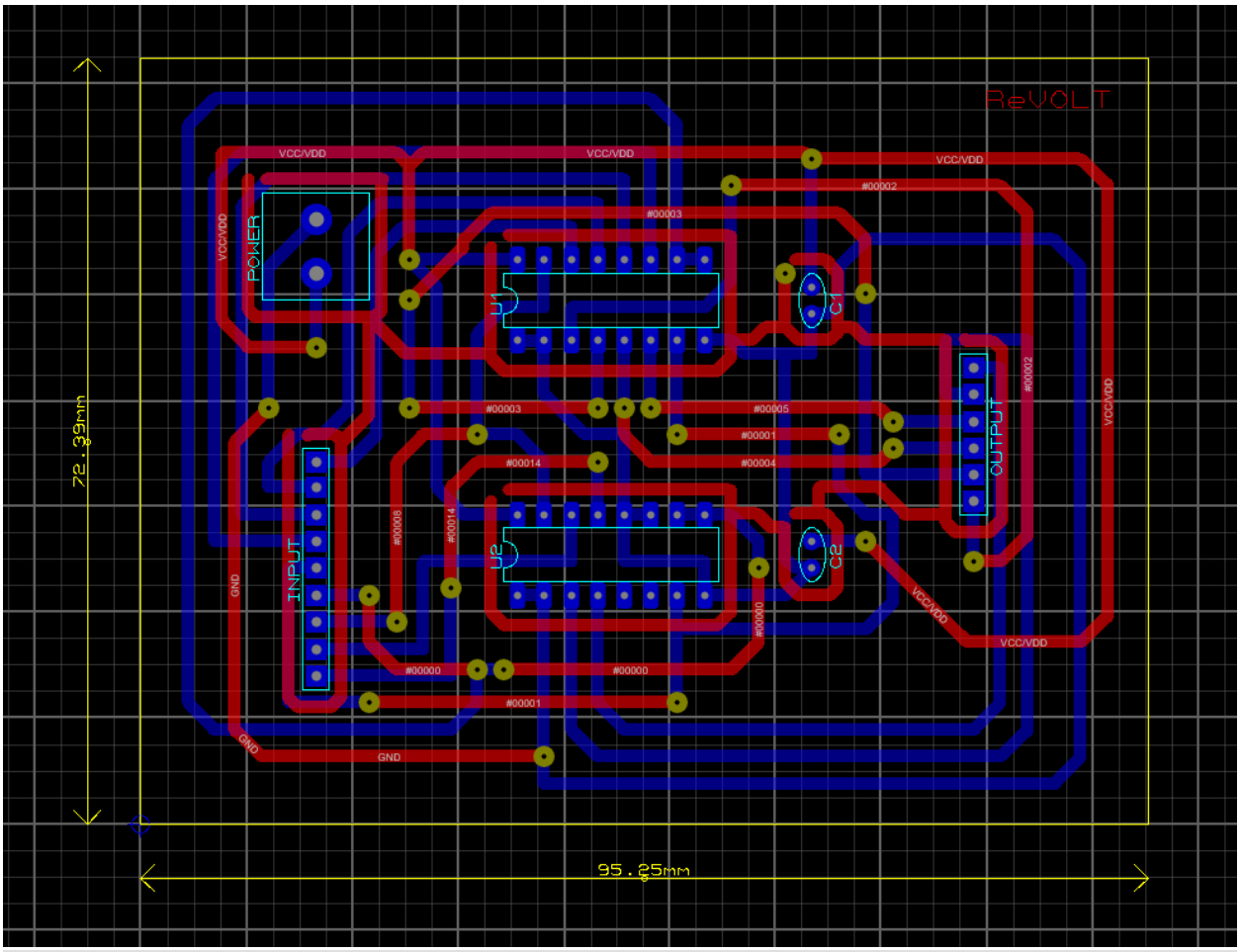
6 outputs are ALU0, ALU1, ALU2, ALU3, ALU4, ALU5.

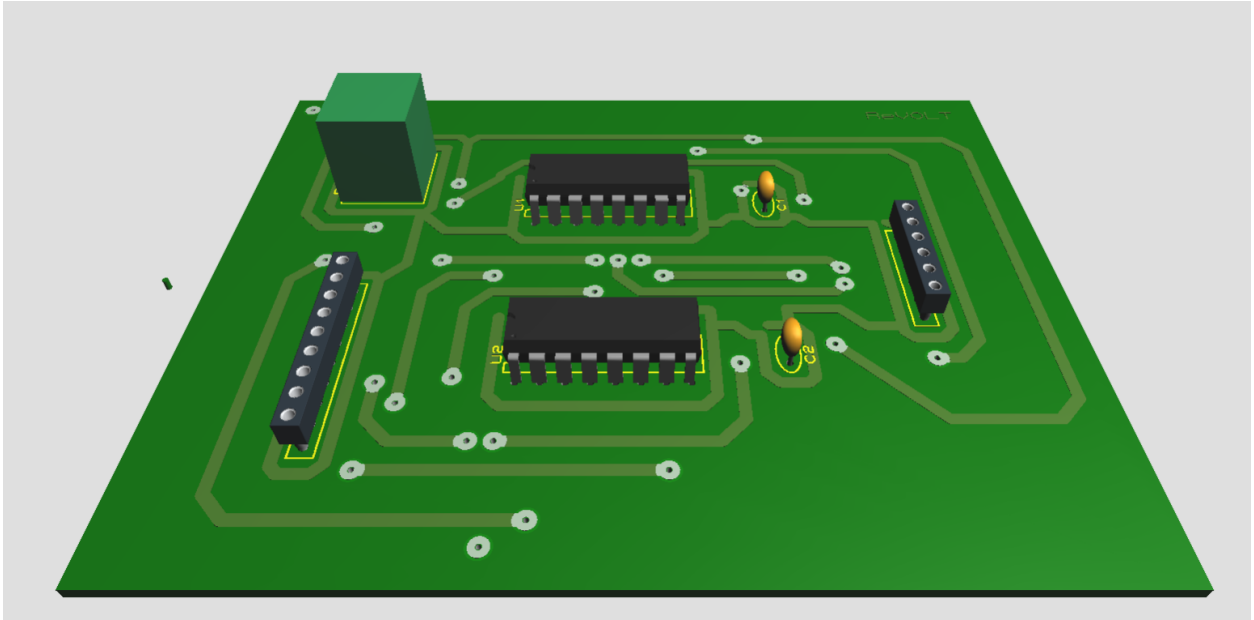
Description of used ICs:

IC 74HC173 – Quad D-type flip-flop positive-edge trigger:

Described earlier.

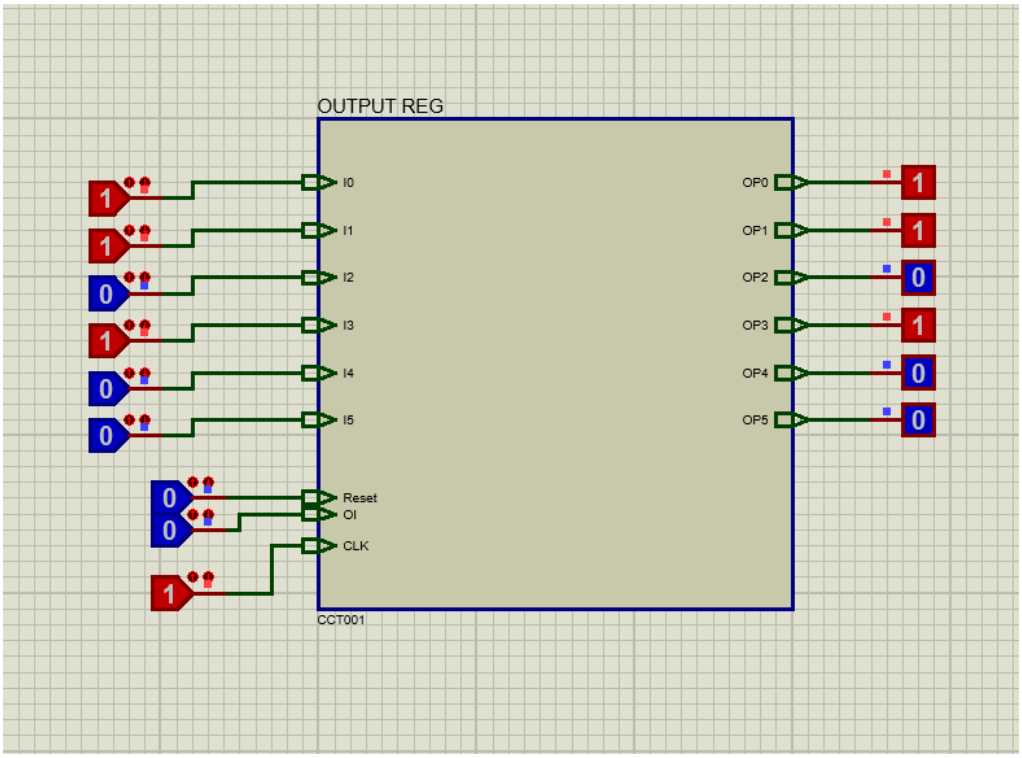
PCB Layout:

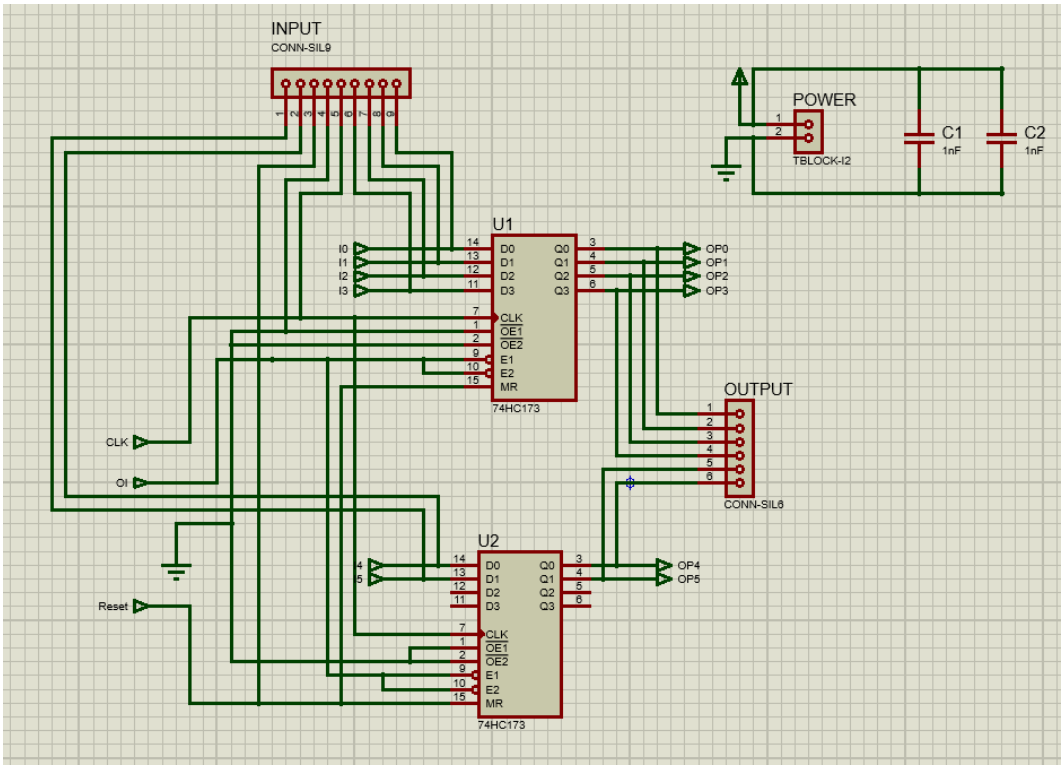

ReVOLT
VCC/VDD
POWER
INPUT
OUTPUT
GND
U1
U2
C1
C2
#00000
#00001
#00002
#00003
#00004
#00005
#00008
#00010
#00014
72.39mm
95.25mm


ReVOLT
U1
U2
C1
C2

Output Register:

Schematic Layout:





Design:

Same as B Register except for the input and output pins.

Input Pins:

OI is active low. When OI=1 the Output Register doesn't take input and output shows the memory and when OBI=0 the output changes with the clock function.

Reset is active high. So if reset=1, the outputs resets to initial value 0.

When the clock turns from 0 to 1, the output is changed according to the input, because of positive edge triggering.

Whether the data will go to output side from the input side in the IC is dependent on the OE1', OE2' pins. These are active low pins, so the were grounded to always pass the data in the IC.

Output Pins:

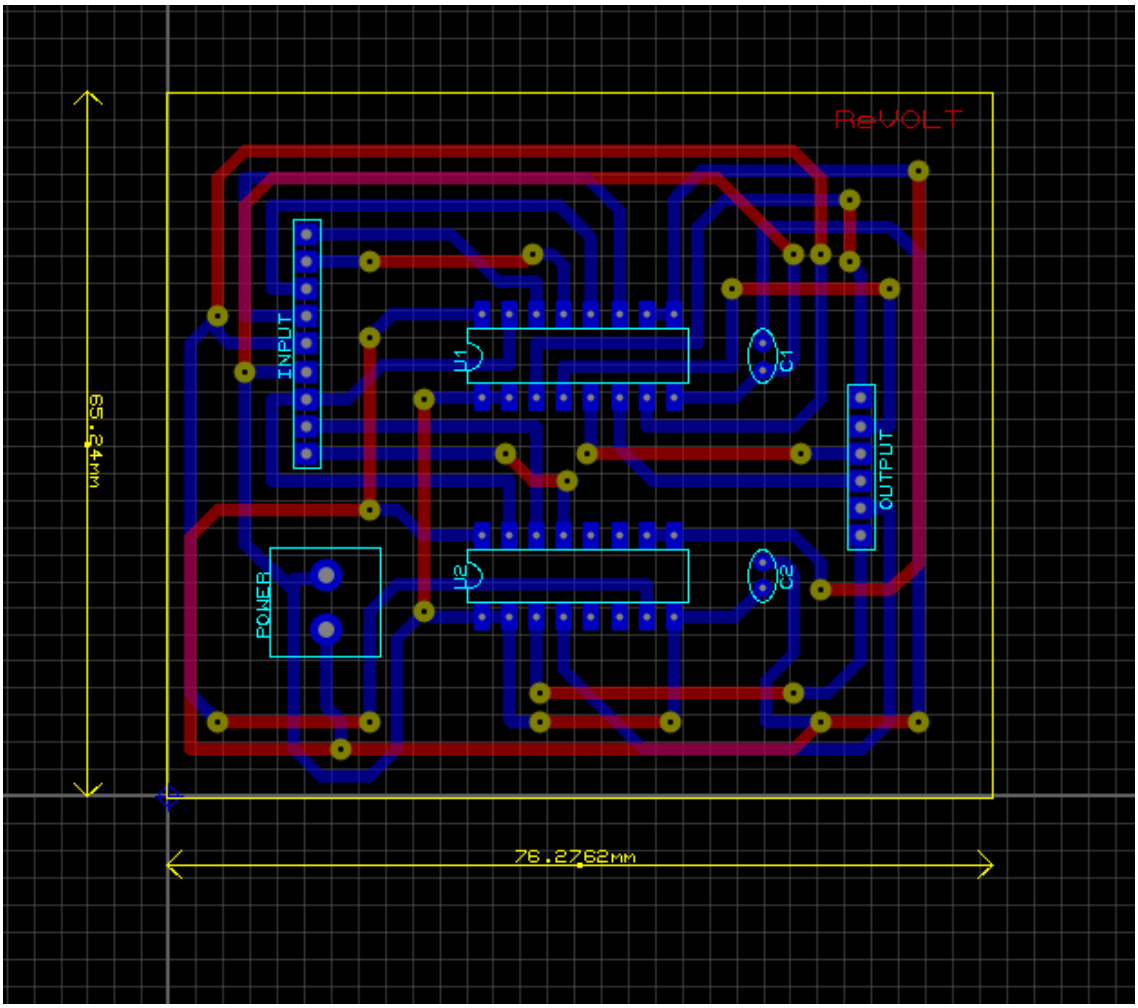
6 outputs are OP0, OP1, OP2, OP3, OP4, OP5.

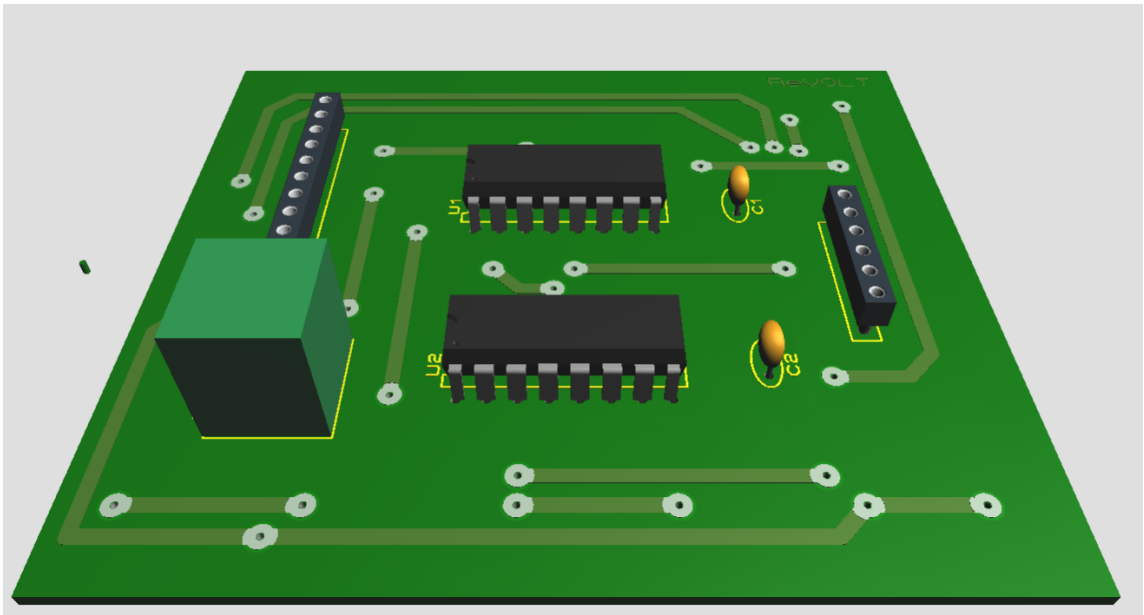
Description of used ICs:

IC 74HC173 – Quad D-type flip-flop positive-edge trigger:

Described earlier.

PCB Layout:


ReVOLT
INPUT
OUTPUT
POWER
U1
U2
C1
C2
65.4mm
76.2762mm

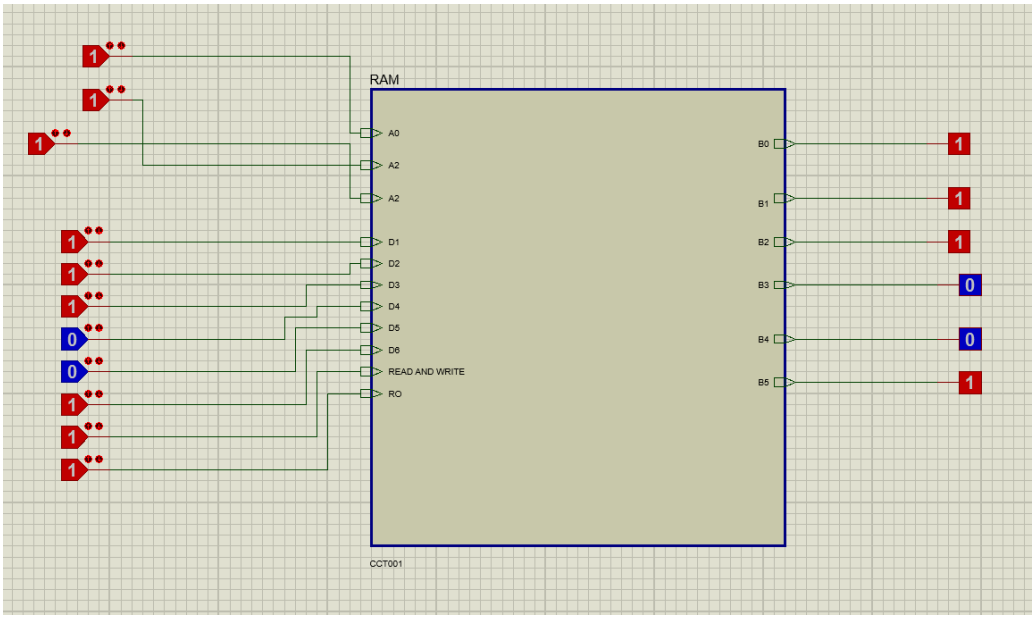

ReVOLT
U1
U2
C1
C2

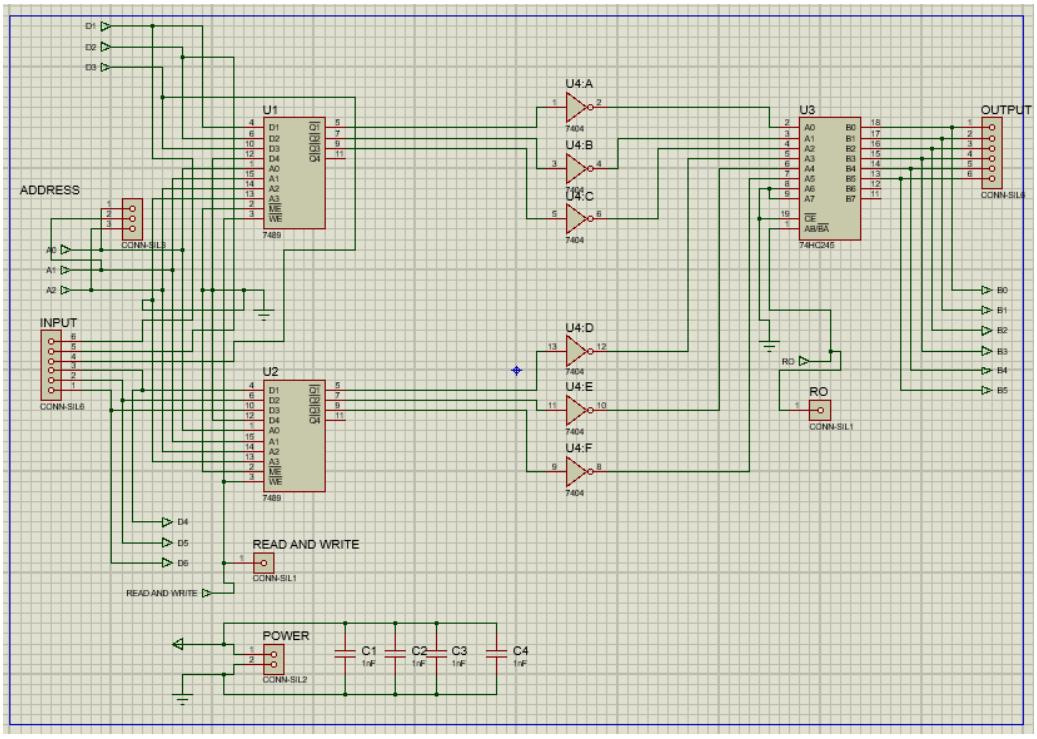
2. RANDOM ADDRESS MEMORY (RAM)

Ram is a 8*6 bit memory .It is actually a memory address register which operates using address switches and input units .

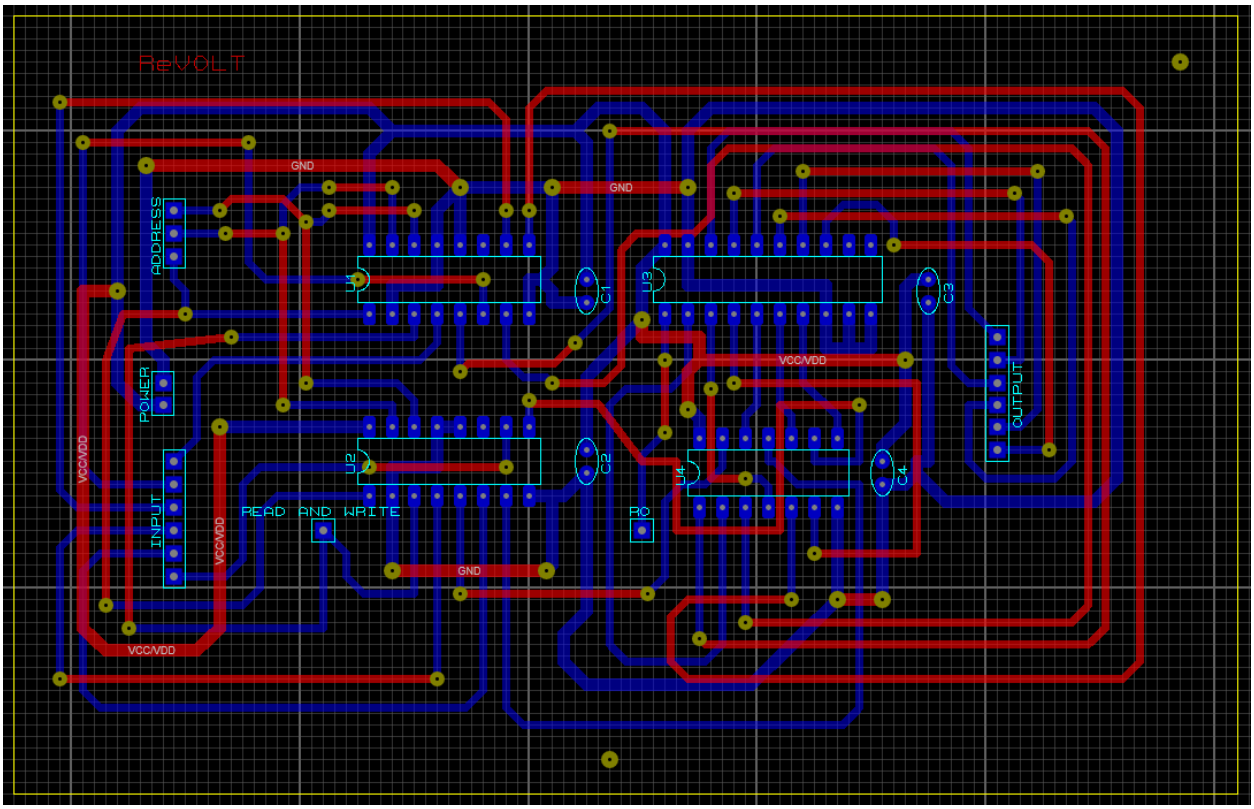
Here two ICs (7489), one 74HC245 and not gate were used .

CIRCUIT DIAGRAM:





PCB Layout:



PIN DESCRIPTION

In 7489 ic , there we used 4,6,10 no pin as input .As we have to build a 6 bit SAP . We used 2 ics . Then for the address selector pin , we used 1,15,14 no pin to put . The 3 no pin of 7489 ic is used as enable pin . Its in active low mode . That means whenever the WE pin is off then RAM will be activated. Also there is a pin named RO , its also an enable pin which helps to active the output . It's a active high pin .

Operation of the RAM

Prog state

When the pin no 3 of the IC 7489 is in High state, then RAM is in Prog Mode. In this mode, the Address selector takes input from the address

pins of the input unit and passes those addresses to the address pins of the RAM. The user, keeping the Write mode on (pin1 of IC 74HC157 Low), gives inputs to the addresses and the inputs are stored in those corresponding addresses

Run State

When the pin no 3 of IC 7489 is in Low state, then the RAM is in Run Mode. In this mode, the address selector takes input from the output of Memory Address Register and sends those addresses to the address pins of the RAM. Here, keeping the Read mode on (pin 1 of IC 74HC157 High), the RAM will show corresponding output to the addresses in which input was stored in Prog Mode. The addresses will come from MAR , and the input stored in those addresses previously in prog state, will be shown as output in Run state.

Contribution:

210021118: Program Counter; Report Making

210021139: Accumulator; PCB of Program Counter, Accumulator, B Register, Output Register; Report Making

210021212: MAR, Input Unit

210021214: MAR, RAM, PCB of MAR, RAM

210021238: Design of Program Counter, Accumulator, B Register, Output Register; Report Making

210021242: Clock Pulse Generator designing & PCB, PCB of Input Unit, Accumulator, B Register, Output Register

210021256: Cross-check